Top Down or Bottom Up?

Evidence from a Field Experiment on Public Goods Provision

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Bottom-up initiatives have become a popular alternative to the traditional top-down provision of local public goods. This study compares the effectiveness of these two approaches. Based on a randomized controlled trial with 120 communities in rural El Salvador, we assess the impact of two interventions addressing solid waste contamination: (i) top-down service delivery through external cleaning teams, and (ii) capacity building for bottom-up provision through local facilitators. Using an objective measure of pollution based on geotagged photos and deep learning, we find large immediate effects for both interventions, with reductions in waste by 50 percent for the top-down intervention and 39 percent for the bottom-up intervention. Despite requiring twice the monetary investment, the bottom-up intervention showed no clear sustainability advantages, with effects diminishing by 47–72 percent for both approaches four months after the withdrawal of support. Our complementary data from 2,421 surveys and 883 activity records is consistent with a theoretical framework where many individuals are willing to contribute to public goods when others do, but fail to coordinate in the absence of a committed leader.

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1 Introduction

Many of the world's most pressing challenges, like curbing greenhouse gas emissions, guaranteeing public safety, or establishing a functioning health and education infrastructure in low-income countries, are public goods problems. As public goods benefit everyone irrespective of their personal contribution to them, individuals have an incentive to free-ride. To avoid the resulting underprovision, the standard solution calls for a *top-down* intervention by a powerful actor such as the state to deliver the public good or enforce rules for its protection (Olson, 1971). Yet, ample empirical evidence documents that groups are often able to act collectively and overcome the social dilemma tied to public goods (e.g., Ostrom, 1990, 1999). This has inspired an alternative line of thinking that advocates for investing in local capacity to enable *bottom-up* provision. Such locally-driven solutions are often hailed to yield more sustainable results by fostering community ownership and reducing dependency on external support.

The notion that development projects should aim at sustainability through local capacity building has become highly influential in development thinking (Kremer and Miguel, 2007). In the last few decades, there has been a marked shift away from large-scale, top-down schemes towards interventions that strengthen local capacity and promote self-help. The appeal is clear: rather than creating dependency on external aid, a one-time investment in local collective action capacity could lead to sustainable public goods provision through enduring behavioral and institutional changes. While evidence on such community-based initiatives has been growing, a key question remains largely unanswered: how effective are such bottom-up approaches compared to the traditional top-down alternatives they seek to replace?

We study this question based on a randomized controlled trial with 120 communities in rural El Salvador. We compare the impact and sustainability of two four-month programs designed to reduce local solid waste pollution. The first intervention pursued direct top-down service delivery with monthly community visits by an external cleaning team to collect litter from the streets. In the second intervention, a local facilitator was appointed for each community to raise awareness and mobilize for collective action to address the problem in a bottom-up process. Typical activities in this capacity-building initiative were educational sessions about waste management, community meetings to define common strategies, and collective monthly cleanups. While a professional waste collection service may be more efficient due to specialization and economies of scale, the bottom-up intervention could achieve greater and more sustained reductions in contamination by discour-

aging littering, fostering social norms around waste disposal, and strengthening local collective action. We randomly assigned communities to three experimental groups: top-down intervention (40 communities), bottom-up intervention (39 communities), and control (41 communities). To track waste contamination levels in all communities, we took approximately 200,000 geo-tagged photos along all public streets, and evaluated them using a deep learning model. Our model achieves performance comparable to leading trash-detection models, allowing us to establish a reliable and objective measure of contamination. To understand the mechanisms behind potential impacts, these contamination assessments were complemented with survey data from a sample of 2,421 residents and a detailed registry of all the 883 activities conducted in the context of the interventions. We estimate effects immediately after the four-month intervention period and again four months after the withdrawal of support to assess sustainability.

We find large *immediate impacts* for both interventions. Top-down service delivery reduced solid waste contamination by 50 percent ($p < 0.01$). Effects are 11 percentage points smaller ($p < 0.1$) for the bottom-up treatment, with a reduction by 39 percent ($p < 0.01$). The bottom-up intervention was also substantially less cost-effective, requiring more than twice the monetary investment and four times the total resources when accounting for volunteer time. Relatedly, the average community cleanup required six times as many worker-hours compared to top-down provision. This low productivity implies considerable time costs for community volunteers, 85 percent of whom were women or children. *Sustainability results* show large and statistically indistinguishable ($p = 0.25$) depreciations in effect sizes for both treatments. Four months after the withdrawal of external support, we observe effect sizes corresponding to 14 percent for the top-down intervention ($p = 0.20$) and 21 percent for the bottom-up intervention ($p < 0.05$), with no statistically significant difference between treatments ($p = 0.47$). Despite substantially higher initial investments, we thus find limited evidence that capacity building for bottom-up provision yields greater sustainability.

Our complementary survey and registry data also allow us to explore the role of two broad channels of bottom-up public goods provision. First, provision can be achieved through shifts in *individual behaviors and social norms*, such as littering and disposal practices. In this context, capacity building can alleviate informational constraints by making people more aware of the problem or of other people's willingness to contribute to its solution. We find limited evidence for such behavioral effects. While the bottom-up intervention generated substantial short-term improvements in recycling and littering norms, these effects were similar in the

top-down treatment and largely disappeared after withdrawal. The only exception are waste disposal practices within households, where only the bottom-up intervention produced immediate and partly sustained reductions in waste burning.

Second, provision can be achieved by coordinating *collective action*, such as community cleanups. Here, capacity building can alleviate organizational constraints by building social capital and local leadership. Our data is consistent with this channel. The bottom-up intervention produced substantial increases in social capital ($p < 0.01$), primarily driven by large effects on volunteering that are likely attributable to participation in cleanup campaigns. Yet, much of this success is tied to the presence of the facilitator. While we observe high and stable levels of collective action during the intervention period (0.9 monthly cleanups per community), this is followed by an immediate and sharp drop in the number of activities after facilitator withdrawal, with participation numbers remaining stable conditional on campaigns taking place. This pattern is consistent with a coordination failure, where residents are willing to contribute to a cleaner community but struggle to overcome the transaction costs of organizing collective action. This coordination problem was temporarily solved through the leadership of the facilitator, but in most communities the intervention failed to establish enduring structures of local organization and leadership.

This study contributes to the literature on public goods provision in poor communities. The rise of bottom-up development strategies focusing on local capacity represents a major trend in development practice (Casey, 2018; Mansuri and Rao, 2012; Kremer and Miguel, 2007). Our analysis of 250,000 World Bank project reports illustrates the scale of this shift away from serving the poor towards empowering them. While terms related to bottom-up provision were almost completely absent from documents in the first few decades of the Bank's existence, by the early 2000s, 60 percent referenced capacity building, 40 percent mentioned community-based development, and 70 percent contained the term sustainability. Our expert survey further shows that practitioners and academics alike tend to be highly optimistic about bottom-up initiatives, with roughly 80 percent of respondents in both groups believing they outperform traditional top-down solutions in the long run. A growing body of empirical research shows that community-based initiatives can successfully deliver public goods, but previous studies have largely focused on comparing bottom-up approaches to a no-intervention baseline rather than to alternative delivery models (Casey, 2018). Our experimental results show that while bottom-up capacity building effectively promoted bottom-up public goods provision, it proved substantially less cost-effective than traditional service delivery,

without offering clear sustainability advantages. This suggests that directly providing public goods – or strengthening state capacity to do so – may outperform investments in parallel bottom-up structures.

Our study also sheds light on a key feature of bottom-up development programs that has received little attention so far: they often do not only shift "power to the people" (Björkman and Svensson, 2009) but also transfer responsibilities. Among all World Bank project reports mentioning capacity building or community-based development, the majority (55%) also reference financial (33%) or labor (40%) counterpart contributions. Despite this, academic and policy focus has almost exclusively been on the power rather than the contribution margin of such programs. A notable exception is Tompsett et al. (2023), who studies the impact of contribution requirements in the context of community well construction in Bangladesh, finding that both monetary and labor contributions decrease cost-effectiveness. Our study aligns with these findings, showing that the costs of bottom-up provision through local voluntary action can be substantial and may thus act as a form of regressive taxation. When designing bottom-up development programs, it is thus essential to consider the extent to which these programs serve and empower their intended beneficiaries rather than compensating for inadequate state service delivery by placing additional demands on poor communities.

We also add to the literature on the management and protection of environmental resources (e.g., Rustagi et al., 2010; Bernedo Del Carpio et al., 2021; Alpízar et al., 2019). A nascent strand within this literature examines interventions to address the growing waste crisis. Humanity generates over 2 billion tons of municipal solid waste annually, and waste generation in low income countries – where less than half the waste is collected – is expected to triple by 2050 (Kaza et al., 2018). Waste contamination poses a serious public goods problem, with negative environmental consequences such as water pollution and soil degradation (Igboama et al., 2022), as well as health risks ranging from infectious disease transmission and respiratory problems (Mohan and Joseph, 2021) to adverse infant health outcomes from marine microplastics (Du et al., 2024). Our results suggest that solutions based on awareness and voluntary local action alone are unlikely to be cost-effective and sustainable. Instead, tackling the problem may require supporting or complementing governments in establishing functional waste collection systems.

Finally, our study advances a growing field of research that leverages machine learning to track and understand global development. While recent work shows that socio-economic outcomes can be predicted from alternative data sources like satellite imagery (Jean et al., 2016; Yeh et al., 2020), phone records, (Blumenstock

et al., 2015) or social media activity (Jakob and Heinrich, 2026), practical applications remain scarce (for a notable exception, see Leeffers, 2026). Our study showcases that waste contamination can be accurately measured using geotagged photos and deep learning. Our automated contamination measure outperforms subjective alternatives, such as enumerator assessments or survey-based perceived contamination, across multiple quality criteria. This highlights the potential of deep learning methods in settings where large amounts of data must be processed or human measurements are highly subjective.

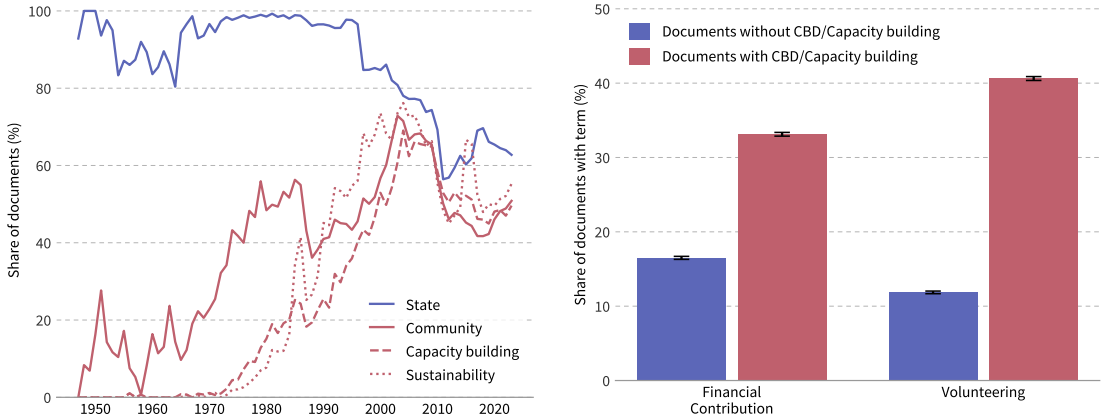
2 The rise of bottom-up approaches to public goods provision

The *public goods problem* models a situation where the benefits of a cooperative outcome accrue to everyone irrespective of people's individual contributions towards it. The dominant strategy for a self-motivated and rational agent is to free-ride by not contributing anything. Standard economic theory considers this a market failure, as it results in a single, Pareto-inefficient equilibrium where the public good is not provided, and individuals fail to realize a mutually beneficial outcome (Olson, 1971; Hardin, 1971, 1982). The conventional approach to addressing market failures associated with public goods calls for a *top-down* intervention by a powerful entity, such as the state, to either supply the public good or enforce regulations for its protection.

However, ample research documents that most people do not behave as the standard model of self-interested actors predicts. Zero contributions to public goods are neither the norm in laboratory experiments (e.g., Fischbacher et al., 2001; Willer, 2009; Chaudhuri, 2011) nor in real-world situations. For example, many people volunteer in associations, donate blood, contribute to charities, make environmentally friendly consumption choices, or take part in political protest. Rather than maximizing personal gains, the majority of individuals appear to follow norms of reciprocity and contribute as long as a sufficient number of others do so too (e.g., Keser and Van Winden, 2000; Gächter, 2006; Thöni and Volk, 2018). Under preferences for conditional cooperation, the provision of public goods becomes a coordination problem with multiple possible equilibria. This is consistent with numerous examples showing that groups sometimes succeed and sometimes fail in providing public goods or protecting common resources *bottom up* (e.g., Ostrom, 1990, 1999).

In this context, the idea has gained traction that groups can be empowered to coordinate and guarantee the provision of public goods in a bottom-up process. Under this model, rather than investing directly in public goods delivery, external actors invest in *local capacity building* to enable communities to address public goods problems through their own efforts. This idea is strongly reflected in approaches such as community-based development (CBD), community-driven development (CDD), or participatory development (Mansuri and Rao, 2012; Casey, 2018).

Figure 1: The rise of bottom-up development strategies



(a) Development terminology over time (b) Contribution requirements by approach

Notes: Illustration based on 263,018 project documents obtained through the World Bank API. We exclude documents with less than 500 correct English words (6% of all documents), and documents that do not contain the word "development" (15% of the remaining documents). Panel (a) shows the share of documents mentioning different development approaches over time. Panel (b) shows the share of documents mentioning terms related to community contributions, comparing documents with CBD/capacity building (red) with those not mentioning these terms (blue). *CBD* includes keywords such as "community-based development", "community-driven development", "participatory development", or "local development"; *Financial contribution* includes "counterpart fund", "matching grant", "co-financing", "cost sharing", or "local contribution"; and *Volunteering* includes "volunteer", "voluntary", "labor contribution", "unpaid", or "sweat equity" (different spellings accounted for).

The *rise of bottom-up strategies* represents a major strategic shift in international development cooperation. Amid concerns about poorly maintained infrastructure following traditional top-down interventions, governments, NGOs and international organizations have increasingly turned to community-based solutions for public goods provision. The appeal rests on the idea of planting a "seed" for long-term sustainability: a one-time investment in local capacity that generates lasting behavioral and institutional changes (e.g., Kremer and Miguel, 2007). Accordingly, this bottom-up approach is often hailed as "more responsive to demands, more in-

clusive, more sustainable, and more cost-effective than traditional centrally led programs" (Dongier et al., 2002), and believed to sustainably transform and strengthen local institutions. To illustrate its popularity, we scraped over 250,000 World Bank project documents published between 1947 and 2023. Figure 1a shows that terms related to bottom-up development strategies, such as "capacity building", "community", and "sustainability", were barely mentioned in the first three decades but increased sharply starting around 1970. By the early 2000s, over 70 percent of documents mentioned at least one of these terms, with the share stabilizing at around 50 percent today. We observe similar trends for related terms such as "community-based development", "participatory", and "empowerment" (Figure A2). Meanwhile, mentions of the term "state", typically associated with top-down schemes, have declined sharply (Figure 1a).[1] This focus on participatory, bottom-up initiatives is also reflected in funding priorities. In 2022, the World Bank alone had 373 ongoing community-based initiatives with more than $40 billion in total lending (World Bank, 2022).

While bottom-up approaches are typically framed in terms of shifting power to communities, in practice they also involve *shifting responsibilities*. Among World Bank documents mentioning capacity building or community-based development (CBD), 55 percent also reference contribution requirements, with 33 percent mentioning terms related to financial contributions and 40 percent to labor contributions (Figure 1b and Figure A2). By comparison, contribution-related terms rarely appear in non-CBD/capacity building documents. Hence, the strategic shift from serving the poor to empowering the poor is inextricably linked to a shift in who bears the costs of public goods provision.

The rise of bottom-up development strategies has also sparked interest in the academic community, leading to a number of *rigorous evaluations* to assess the effectiveness of the approach. Table A15 provides a comprehensive overview of this literature. Although the reviewed studies vary in the types of interventions and outcomes they examine, we can draw two general conclusions from this research. First, bottom-up initiatives are indeed often successful in delivering and maintaining public goods and improving the livelihoods of the poor (Avdeenko and Gilligan, 2015; Björkman and Svensson, 2009; Desai and Olofsgård, 2019; Duflo et al., 2015). Second, the evidence is inconclusive on the proposed transformative impact on local institutions. Many evaluations report no lasting effects on collective action capacity (Casey et al., 2012; Casey, 2018; Mansuri and Rao, 2012) or the empowerment of

[1]These terms may also be used in other contexts, such as environmental sustainability or state capacity building, and are therefore imperfect proxies for the prevalence of bottom-up development approaches.

minority groups (Casey et al., 2012; Van der Windt and Mvukiyehe, 2020). Yet three key limitations remain. First, existing studies typically compare bottom-up initiatives with a status quo where no infusion of funds occurs. As Casey (2018) notes in a recent meta-study, "remaining questions are versions of the perennial *compared to what* challenge. While [community-based development] appears to effectively deliver local public goods, the formal state could also be assigned this task." Second, there is limited understanding of the mechanisms through which bottom-up interventions should affect public goods provision. Third, with few exceptions (e.g., Tompsett et al., 2023), research focuses on the power margin of such programs rather than the responsibility margin, neglecting the labor and financial contributions often required from communities. Our study addresses all three limitations by (i) directly comparing capacity building for bottom-up provision with traditional top-down service delivery, (ii) examining the underlying mechanisms, and (iii) studying costs and cost bearing.

3 Research design

3.1 Context

We conducted our study in the context of solid waste management in rural communities in El Salvador, a lower-middle-income country in Central America. Inadequate waste management is ubiquitous in developing countries and causes numerous detrimental health and environmental impacts. Over time, waste can spread over large areas, contaminating rivers, oceans, groundwater and soil. In addition to its environmental consequences, contaminated water can pose serious health risks by spreading infectious diseases such as diarrhea or hepatitis (Mohan and Joseph, 2021). While 96 percent of solid waste in high-income countries is collected and properly disposed of, the corresponding figures are only 51 percent for lower-middle-income countries and 39 percent for low-income countries (Kaza et al., 2018). At the same time, solid waste generation in low- and middle-income countries is expected to triple by 2050. Finding out what works to address the problem is thus an important and timely priority (see Table A16 for a review of the emerging literature in this area).

Solid waste contamination poses a typical public goods problem for local communities, as a clean environment benefits everyone regardless of their personal contribution towards it. This is in line with insights from our baseline survey, showing that even though most people in our sample are bothered by the waste pollution

in their communities, the problem remains widespread. Nearly 80 percent of respondents indicate that the waste in their community bothers them much (39%) or very much (38%). Meanwhile, about half the people admit that they dispose of their waste improperly by burning, burying, or dumping it. Similarly, only 32 percent of the communities are visited by a municipal garbage truck collecting household waste at least every two weeks, and 36 percent have no such truck service at all. Almost all communities have considerable amounts of waste in public spaces, and contamination levels are not significantly correlated with the frequency of the garbage truck service.

3.2 Treatments

Two distinct strategies exist to tackle the problem. People can either stop dumping waste in public spaces, or waste can be removed and properly disposed of. Since both actions involve costs, and gains are shared between all residents, effective solutions are needed to overcome the free-riding problem inherent to the provision of public goods. We partnered with the local NGO Consciente to develop and implement two four-month interventions to address the problem: (i) a traditional top-down service delivery intervention and (ii) a capacity-building intervention for bottom-up provision.

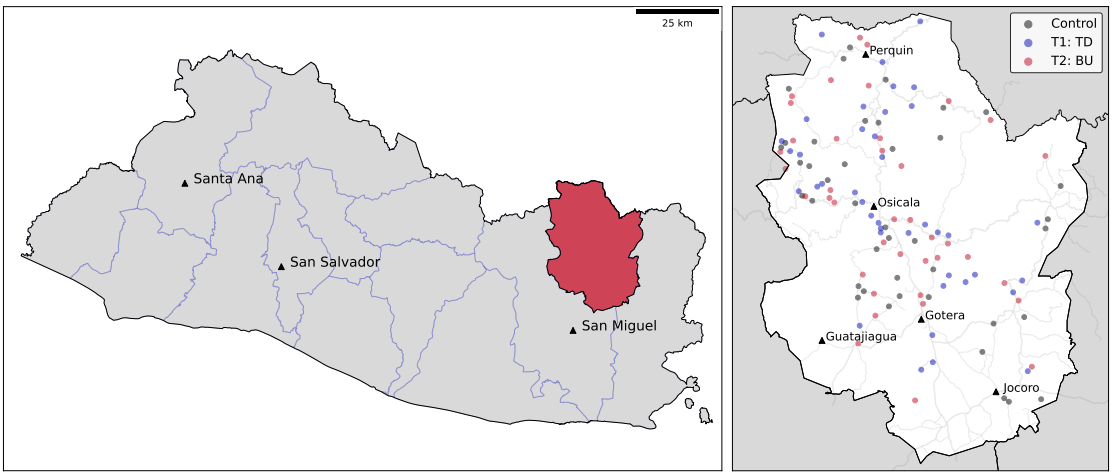
In the *top-down intervention*, an external team of cleaners employed by the NGO made monthly visits to all communities to collect litter from public areas and gather household waste from residents. The team comprised two cleaners and a garbage truck driver, and each community visit typically lasted half a day. While the intervention was conducted by a non-governmental rather than a governmental institution, it mirrors what a top-down state intervention would look like in this context and corresponds to the typical approach pursued by governments worldwide to tackle solid waste pollution. Implementation was consistent across communities, with an average of 3.9 cleaning visits per community (roughly one per month) and no community receiving fewer than 3 visits.

In the *bottom-up intervention*, a team of 24 part-time facilitators was hired and trained in topics related to waste pollution and management and community organization strategies. Facilitators were typically young university graduates from the area, but not necessarily from the community, and responsible for one or two communities. Their job consisted in building local capacity to address the waste problem by raising awareness, sharing knowledge, mobilizing for collective action, and strengthening organizational structures to facilitate sustainable solutions. For this

purpose, they could draw on extensive teaching materials developed by the NGO, but were instructed to adapt the proposed activities based on local needs. The typical community intervention consisted of an initial meeting, a series of educational sessions and community activities on waste management (such as input and discussion sessions, hands-on workshops, poster campaigns, or community movie nights), and monthly collective cleanups. The facilitator also assisted the community in organizing the disposal of the waste collected from the cleanups and households. This was typically done by using the private vehicle of a community member or by appealing to the municipal government. At the end of the intervention, each community presented a waste management plan indicating how the problem would be addressed after the withdrawal of the NGO. The capacity-building activities were successfully implemented in 95 percent of assigned communities, with an average of 18.6 activities per community (15.2 capacity-building activities and 3.4 collective action activities, see Table A11). About 20 community members, representing an estimated 7 percent of the population, participated in a typical activity. For simplicity, we refer to this as the bottom-up intervention throughout, although the treatment itself consists of capacity building for bottom-up provision.

To gauge expectations about these interventions, we conducted an expert survey with 100 scientists and practitioners (Figure A1). Results are consistent with the sense of optimism regarding bottom-up strategies discussed in Section 2. Over 90 percent predicted that the bottom-up intervention would reduce waste contamination and littering in both the short and long term, and about 80 percent expected it to outperform the top-down intervention in the long run. While the majority of academics believed that the relative advantage of the bottom-up approach would unfold only in the long term, most practitioners also predicted better short-term outcomes.

Figure 2: Study area



Notes: Left panel shows El Salvador with the department of Morazán highlighted. Right panel shows the locations of the 120 study communities by treatment assignment.

3.3 Sampling and randomization

To study the impact of the two treatments, we conducted a randomized controlled trial with 120 communities in the rural department of Morazán in El Salvador (see Figure 2). The selection of these communities was undertaken in two steps. First, we compiled a list of medium-sized, non-urban communities (30–300 households) facing waste management problems with the help of municipal governments. In a second step, we conducted a baseline survey in 140 communities and selected 120 communities based on two criteria: moderate or high contamination levels, and spatial distance (to limit spillover effects). The resulting sample is not representative of our study area, but contains a diverse set of communities that have not solved the waste management problem either through an endogenous bottom-up process or with the help of local government institutions. We randomly assigned these communities to three experimental conditions: (1) the top-down intervention (40 communities), (2) the bottom-up intervention (39 communities), or (3) a control group that received no intervention (41 communities).[2] Randomization was stratified by baseline contamination (three bins) and geographic zones (four bins). In all experimental groups, we conducted a measurement wave before the intervention (baseline), toward the end of the intervention (midline, after 3–4 in-

[2]The number of communities differs between experimental groups because remainders per stratum were assigned with probability 1/3 to each group or group combination (in the case of two remainders). A common alternative is to group remainders over all strata and reassign them randomly. Assigning remainders with probabilities instead of grouping them means equal group sizes cannot be ensured, but assignment balance within the strata is preserved (McKenzie and Bruhn, 2011).

tervention months), and four months after the intervention (endline). This allows us to study both the immediate impact of the two provision models and whether potential effects are sustained after the end of the program.

3.4 Data

To track different waste-related outcomes, we collected three types of data: (i) contamination assessments based on images taken along all streets, (ii) survey data on people's perceptions and self-reported behavior, and (iii) monitoring data on all the activities that were conducted in the context of the interventions.

3.4.1 Image data on contamination

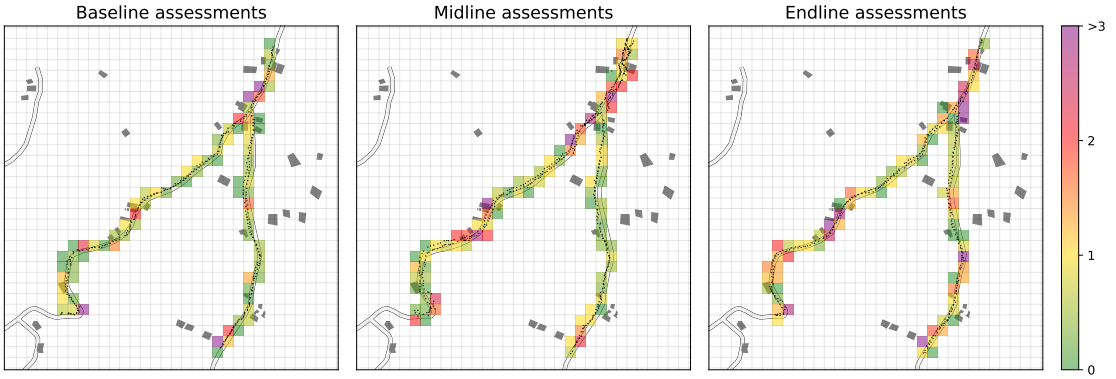
For the main outcome of our experiment, we took *photos* along all the streets and public spaces in the 120 communities. For this purpose, enumerators worked in pairs and simultaneously took geotagged photos on both sides of the street every five steps. Enumerators were carefully trained and received a detailed manual explaining how to take the photos. Photos typically show a portion of the street, the roadside, and the background. To ensure spatial consistency across the three measurement waves, we used an application that enabled us to outline the geographic boundaries of each community and display them on an interactive map. Enumerators were instructed to cover all roads, paths and public spaces within this designated area. To account for minor deviations in the covered area, we only include photos with spatial support across all three waves (92% of all images).[3] This procedure results in approximately 500 images per community and wave, ranging from 118 photos in the smallest community to 1,926 photos in the largest community, and a total of 181,393 images across all waves. We then used a deep learning model to predict the number of trash pieces on each image (see Section 3.5).

A key challenge is linking midline and endline contamination levels in different areas of each community to their baseline contamination values. We use two *spatial aggregation* approaches to address this challenge: raster and kernel aggregation. For our main pre-registered specification we use *raster aggregation*, where we overlay a fixed 16.5 x 16.5 meter grid on each community and compute wave-specific averages

[3] A photo in a given wave is defined as having no spatial support in another wave if the closest photo is more than 8 meters away and the fifth closest photo is more than 25 meters away. This decision rule was found to produce good results, by excluding road segments that were not covered in all waves, but keeping photos in all other segments. We only include photos with spatial support in all waves, e.g., only baseline photos with nearby midline and endline photos.

across all photos in each cell (Figure 3).[4] Each cell contains 4.6 images on average, and 81 percent of cells with baseline images also contain both midline and endline images. This yields a final sample of 10,260 cells, averaging 90 cells per community and ranging from 21 cells in the smallest community to 278 in the largest. As a robustness check, we use *kernel aggregation*, drawing a 12.5 meter radius circle around each midline or endline image (Figure 4) and computing weighted averages of baseline contamination values using a triangular kernel.[5] The average circle contains 7.6 baseline images, with 99 percent containing at least one baseline image. This produces 60,702 observations for immediate effects (midline) and 65,671 for long-term effects (endline).[6]

Figure 3: Illustration of raster approach in example community



Notes: Black dots represent image locations. Cell color corresponds to the average number of trash pieces identified on images in the cell. Resolution of the raster is 0.00015 degrees (approx. 16.5 m). We use OpenStreetMap for all base maps.

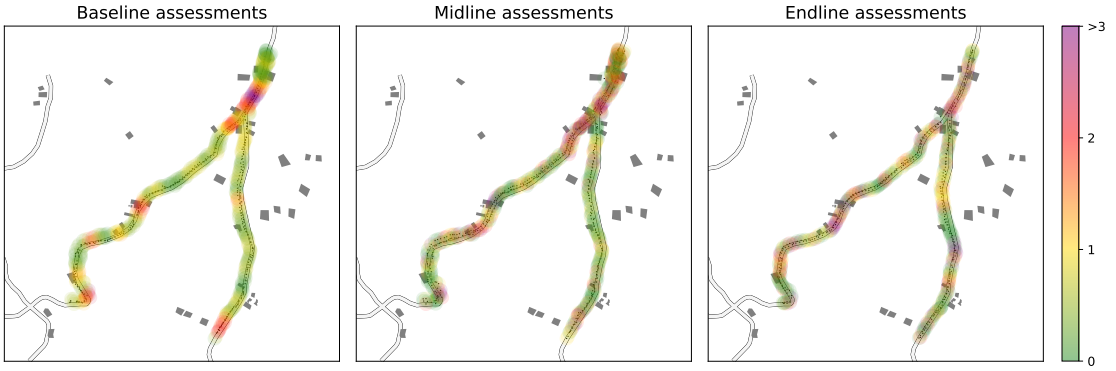
To assess the robustness of our objective deep-learning derived measure (see Section 3.5), we also collected data on three more *subjective contamination measures*. First, three independent enumerators manually classified a subset of 50 photos per community and wave into four categories, ranging from "very clean" to "very dirty". All photos were classified by all three enumerators in random order to prevent systematic biases, and we take the mean across the three assessments. We use raster aggregation with a 0.0006 degree grid (approximately 66 x 66 meters) and kernel aggregation with a 50 meter radius for this photo assessment data, yielding 1,897

[4]The grid size balances two competing criteria: generating sufficient observations per community while maintaining good spatial support across waves to minimize dropped cells. A 16.5 x 16.5 meter grid strikes an optimal balance.

[5]This approach creates different baseline circles for midline and endline measurements respectively. We selected the 12.5 meter radius by examining community maps with various circle sizes; this radius ensures that nearly all dropped observations occur at community boundaries rather than within communities.

[6]As a simple additional robustness check, we also compute raw community averages across all images for each community and wave, yielding 120 observations.

Figure 4: Illustration of kernel approach in example community



Notes: Black dots represent image locations. Circle color corresponds to the number of trash pieces identified on each image. Baseline values are imputed based on circles around each midline and endline assessment respectively. Circle radius is 12.5 m. A triangular kernel is used to give higher weights to closer assessments.

raster cells with support across all waves and 12,400 kernel observations. Second, enumerators made on-site subjective assessments of the general cleanliness of the environment every 25 steps or 5 photos. Based on representative example images provided during training, they classified their environment into four categories, again ranging from "very clean" to "very dirty". Our final sample consists of about 100 geo-coded ratings per community and wave, with 23 assessments in the smallest community and 408 in the largest community.[7] For this on-site assessment data, we use raster aggregation with a 0.0003 degree grid (approximately 33 x 33 meters) and kernel aggregation with a 25 meter radius, yielding 4,074 raster cells with support across all waves and 25,595 kernel observations. Third, our survey includes questions on perceived community cleanliness, allowing us to construct measures of perceived contamination at the community level (see Section 3.4.2).

We use three quality criteria to assess *reliability* of our measures: (i) temporal stability, measured by the inter-wave correlation in community-level averages (higher is better); (ii) enumerator dependence, measured by the intra-class correlation (ICC) of enumerator identity (lower is better); and (iii) within-community consistency, measured by the average absolute standardized difference between simultaneous measurements on opposite street sides (lower is better). Across all measures, our objective deep-learning derived measure performs best, with the highest inter-

[7]To obtain geocoded ratings, we used a simple low-tech strategy. Enumerators had to take a picture of a placard with the number corresponding to the level of contamination. We then used the weights of a GitHub model pretrained on the popular Street View House Numbers (SVHN) dataset (Netzer et al., 2011) to predict the number corresponding to each image. To make sure that all predictions were correct, we manually reviewed the few cases where the model predicted low certainties. The number images were integrated with all other photos to determine the spatial support across waves.

Table 1: Comparison of contamination measure reliability

Measure	Inter-wave correlation	Enumerator ICC	Side difference
Photo trash detection (deep learning)	0.749	0.009	0.363
Photo assessment (enumerator)	0.640	0.012	0.534
On-site assessment (enumerator)	0.461	0.074	0.662
Perceived contamination (survey)	0.381	0.074	-

Notes: Inter-wave correlation shows the average temporal stability between waves (higher is better). Enumerator ICC is the proportion of variance due to enumerator identity (lower is better). Side difference capture the average absolute standardized difference between simultaneous measurements on opposite street sides (lower is better). Inter-wave correlations and side differences are computed at the community level; enumerator ICCs at the level of individual photos or survey respondents. For photo-based measures, the enumerator corresponds to the person who took the photo.

wave correlation ($r = 0.749$) and the smallest enumerator effects ($ICC = 0.009$) and street side differences (0.363 standard deviations). By contrast, survey-based perceived contamination (large enumerator effects and low inter-wave correlation) and on-site assessments (large enumerator effects) are the noisiest measures (Table 1). In addition to these empirical quality advantages, our automated measure may also be less susceptible to systematic bias by treatment status. Because the deep learning model applies the same procedure to all photos identically regardless of treatment assignment, it cannot be influenced by respondents' differential awareness of or involvement in intervention activities – a concern that might affect self-reported survey measures, particularly from the bottom-up intervention.

3.4.2 Survey data

To better understand the mechanisms behind potential effects, we administered short surveys to 20 residents per community. Our survey includes questions on two key channels through which the interventions may operate: (i) *waste-related behaviors and social norms*, including waste disposal practices, recycling, and littering norms; and (ii) *social capital*, including trust, social networks, organization membership, and volunteering. We also collected data on treatment-related activities respondents observed or participated in, as well as socioeconomic background and community characteristics. Table A14 provides an overview of all included survey questions. Participants were selected by enumerators during the community visit for the baseline assessments. Enumerators were instructed to recruit survey participants by randomly knocking on doors until the target of 20 interviews was reached. While the resulting sample is not representative (mainly due to differ-

ent propensities to be home during the day), it is very diverse and comparable across experimental groups (see Section 3.6). Our final sample consists of 2,421 individuals.[8] Attrition was 15 percent in the midline and 24 percent in the endline assessment, resulting in 2,066 observations to estimate immediate effects and 1,832 observations for long-term effects. We find no indication of differential attrition by treatment status (see Tables A4, A2, and A3). Missing values were imputed using the mean of the respective experimental group.[9]

3.4.3 Activity registry

To track implementation and collective action dynamics, a comprehensive registry of all activities performed under each intervention was compiled. For the top-down intervention, we collected data on every cleaning visit, including the amount of garbage collected and the number of worker-hours devoted to the task. The activity registry for the bottom-up intervention contained information about the type and duration of each activity, the number of participants, facilitator preparation time, and subjective ratings regarding activity success and participant interest. For cleanup campaigns, the log additionally recorded how much litter was collected in how many worker-hours, and how waste removal was organized. All intervention activities were registered by the NGO staff responsible for conducting each activity (i.e., cleaners or facilitators). Staff were instructed to report honestly on all activities, and neither pay nor promotion was contingent on the successful execution of these activities. In addition, facilitators were required to submit photos of each activity to the project coordination team of the NGO. For the bottom-up intervention, we also recorded all activities during the post-intervention period through phone calls to community leaders, allowing us to study the extent to which collective action efforts continued after the withdrawal of the NGO. To better understand the challenges communities faced in the post-intervention period, we also conducted interviews with the person who remained in charge in each community after the end of the intervention.

[8]This sample is larger than 2,400 because we grouped 8 communities into 4 community clusters before randomization to avoid spillovers due to geographic proximity (meaning that our analyses include 124 communities and 120 community clusters). The community clusters received the same treatment, but 40 interviews were conducted instead of 20. Throughout the study, these clusters are referred to as communities. Note that for safety reasons, surveys were conducted during daytime hours only.

[9]Missing values were rare, with fewer than 1 percent missings in all our main survey variables presented in Table 4.

3.5 Deep learning for waste detection

We employ a novel approach that uses deep learning to create an objective measure of contamination based on the approximately 200,000 images included in our analysis. This is achieved by fine-tuning a YOLO26 object detection model using publicly available trash datasets and manually labeled photos from our own study. The YOLO26 model is part of the YOLO (You Only Look Once) family, which comprises state-of-the-art object detection systems employed in real-time tasks for robotics, self-driving cars, and video surveillance applications (Terven and Cordova-Esparza, 2023). YOLO26 was released by Ultralytics in January 2026. Ultralytics offers five different model sizes, varying in features such as their mean average precision on the popular COCO dataset (200,000 annotated images) and the number of parameters the model has to estimate (ranging from 2.4 million for the smallest and 55.7 million parameters for the largest model). To balance speed, accuracy and necessary computational power, we opted for the median model, YOLO26m, with an mAP50-95 of 53.1 percent for the COCO dataset and 20.4 million estimated parameters.[10]

We fine-tune our model in two stages. In the first stage, we train the model on several complementary trash-detection datasets. We begin with the publicly available TACO (Trash Annotations in Context) dataset, consisting of 1,500 official images with 4,784 annotated trash bounding boxes (Proença and Simões, 2020). The TACO data often serves as the benchmark dataset to compare the performance of different trash detection algorithms. In addition to the official images, TACO also contains a set of photos with crowd-sourced annotations, which have not yet been subjected to a quality check. We manually reviewed all these unofficial images to exclude instances with incorrect bounding boxes, resulting in 3,432 additional images with 7,511 additional annotations, and a total of roughly 5,000 and 12,000 annotations for the extended TACO dataset (official + unofficial TACO). We further augment the training data with the PlastOPol dataset, another widely used trash-detection dataset comprising 2,418 images and 5,300 annotations (see Córdova et al., 2022). Finally, we also include images from our own data: 600 manually annotated images with 3,024 annotations (200 per wave) and 216 images without any trash.

In the second stage, we further fine-tune the resulting model at a higher image resolution, increasing the input size from the default 640 pixels to 960 pixels. The

[10]The mAP (mean average precision) corresponds to the mean of the average precision (AP) over all classes and IoU (Intersection over Union) thresholds from 0.5 to 0.95 (see below for an explanation).

higher resolution is particularly useful in our setting because many pieces of trash occupy only a small portion of the image and may therefore be difficult to detect at lower resolutions. We consider two variants of this second-stage fine-tuning procedure. In the first, we retrain the model on the full set of images at 960-pixel resolution. In the second, we replace our own full images with four overlapping crops of each image, while keeping the other training data unchanged. The crops overlap by 10 percent to reduce the risk of cutting small objects at the boundaries and provide the model with a more detailed view of small trash items.

Figure 5: Illustration of deep learning model performance



Notes: The image shows model predictions for an example image. The decimal number represents the confidence of the model.

We train our model using 70 percent of the data for training and 30 percent for testing, and compute separate performance statistics for each data source. For training, we use up to 200 epochs and a batch size of 8 in the first stage, and limit the second stage, which uses higher-resolution images, to six hours with a batch size of 4 and early stopping after 20 epochs without improvement. These choices are mainly determined by computational constraints. For prediction, we set the detection threshold to 50 percent, meaning that objects are only detected if the model is at least 50 percent confident of its prediction. Our principal performance statistic is the AP (Average Precision), a measure widely used in the deep learning literature to compare results across different models. This metric is based on the area under the precision-recall curve and thus captures how well the model performs across different certainty thresholds. Following standard practice in the field, we

use AP50, meaning that a predicted bounding box is considered accurate if the intersection between the true and the predicted box corresponds to at least 50 percent of the union of the two boxes. As additional, more intuitive measures, we also report the precision (the proportion of detected instances that are correct), the recall (the proportion of true instances that are detected), and the F1 score (a combination of precision and recall), evaluated at the confidence threshold that maximizes the F1 score.

For the TACO dataset, the model achieves an AP50 of 53.2 percent when trained at an image size of 640 pixels, compared with 55.9 and 55.6 percent, respectively, for the two specifications trained at the higher resolution of 960 pixels. As shown in Table 2, these results are broadly comparable to the best-performing models reported in the literature, which achieve AP50 values ranging from 57.4 percent (Das et al., 2023) to 63.3 percent (Córdova et al., 2022). Performance on our own photos is similar. The overlapping-split specification achieves an AP50 of 54.1 percent on our data, compared with 55.6 percent on TACO. Because this specification performs slightly better on our own photos than the no-split specification, we use it as our preferred model, despite the very small reduction in performance on TACO. Using standard object-detection evaluation, this specification attains a precision of 60.6 percent and a recall of 51.9 percent on our photos. For the predictions used in our empirical analysis, we impose a confidence threshold of 50 percent; at this threshold, precision increases to 83.7 percent, while recall falls to 27.9 percent, suggesting that our model produces few incorrect detections, but misses many true instances. This is consistent with the visual characteristics of the images: many pieces of trash are small, partially occluded, or located in the background, making them difficult to identify even upon manual inspection.

The fact that our model is not perfectly accurate at detecting trash has a predictable impact on treatment effect estimates. First, we know that a non-negligible fraction of all detections are *false positives* due to a tendency of our model to identify other objects, typically stones or leaves, as trash. In our test set, we observe an average of 0.154 false positives per image (38 false positives for 246 images in the test set). Assuming that the number of false positives is unrelated to the treatment status, this implies that the average trash count in all experimental groups is biased upward by 0.154 pieces of trash. This does not, however, affect (absolute) treatment effects, as the bias cancels out when comparing different experimental groups. A second bias is related to *false negatives*. The recall of 0.28 (at a confidence threshold of 50 percent) suggests that our model misses approximately 70 percent of trash on our images (i.e., the false negative rate is 0.72). Assuming that the capacity of

Table 2: Model performance

	Our photos				*TACO*			
	AP50	Precision	Recall	F1	AP50	Precision	Recall	F1
Our model								
Base model (640)	45.0	55.3	45.7	50.0	53.2	72.7	45.5	55.9
No split (960)	52.6	59.1	51.9	55.3	55.9	69.7	50.9	58.8
Overlapping split (960)	54.1	60.6	51.9	55.9	55.6	68.1	50.7	58.1
Other models								
Córdova et al. (2022)	-	-	-	-	63.3	48.4	66.4	56.0
Das et al. (2023)	-	-	-	-	57.4	82.8	49.1	61.6
Majchrowska et al. (2022)	-	-	-	-	62.4	-	-	-

Majchrowska et al. (2022) included the extended TACO dataset in their performance evaluation.

the model to detect a given trash piece is unrelated to the treatment status, the average reported trash count for each experimental group thus corresponds to only 28 percent of the true trash count. Consequently, the treatment effect, reported in pieces of trash, is underestimated by the same factor. As the reduced differences between treatment groups are accompanied by a lower variance, this bias disappears when effects are reported in standard deviations. In summary, under plausible assumptions, raw group means and treatment effects will be biased due to the occurrence of false positives and false negatives, while standardized effects are not. When reporting on group means or effects in pieces of trash (or percent), we will thus present results accounting for these two biases, using the following correction:

$$Y_g = (\hat{Y}_g - FP) \cdot \frac{1}{recall} \tag{1}$$

where Y_g is the true average trash count in treatment group g after applying the bias correction to the predicted trash count $\hat{Y}_g$, FP is the average number of false positives per image in our test set and thus 0.154, and $recall$ is the overall share of true trash pieces that are correctly detected in our test set at a confidence threshold of 50 percent and thus 0.28.[11]

[11]To correct the bias in (non-standardized) treatment effects, we only need to multiply the raw treatment effect by $\frac{1}{recall}$, since the first part of the equation cancels out. Note further that

3.6 Baseline characteristics

Our deep learning model detects approximately 0.6 pieces of garbage in the average image. Based on Equation 1 in Section 3.5, this implies that an average image contains about 1.6 real pieces of trash. Considering that the photos were taken randomly along all streets and not specifically in places with garbage, this indicates substantial solid waste contamination. For the average community, it corresponds to roughly 785 visible pieces of trash on our images alone. There is considerable variance between communities with 0.09 detections (hardly any real pieces) in the average image in the least polluted community and 2.2 detections (7.32 real pieces) in the most polluted community. Similarly, enumerators rated the average site across all communities as a 2 ("a bit polluted") on a scale from 1 to 4. Community averages based on these subjective enumerator assessments range from 1.37 in the cleanest community to 2.78 in the dirtiest community. Contamination tends to be higher in areas with more human activity, such as on major streets or close to buildings, and in communities that are more urban and have lower social capital (Figure A3).

The average survey respondent is 43 years old, and 75 percent of respondents are female. About two thirds of the individuals in our sample have not completed any educational degree (no schooling: 20%, incomplete primary: 45%), 11 percent have a primary degree, 19 percent have completed high school, and 5 percent possess a tertiary degree. On average, respondents believe that roughly 60 percent of people in their community litter, that 70 percent of people in their community disapprove of littering, and that 55 percent of people in their community would punish litterers with a disapproving gesture. The average community has about 300 residents, corresponding to roughly 100 households. People tend to know each other, with the average person reporting that 70 percent of community members are known and 40 percent are friends or family. Approximately 20 percent of respondents belong to a community organization and 30 percent report having done voluntary work for the community in the last month.

Table A1 shows that contamination levels and survey responses at baseline are

the assumptions that the probability of false positives and false negatives is unrelated to the treatment status is likely to be only approximately true. In the case of false positives, one could argue that false detections are more likely in cleaner images (where less space is covered by trash). This would introduce an additional downward bias in treatment effects, as contamination in the (cleaner) treatment group is overstated more strongly compared to the (dirtier) control group. In this case, our corrected treatment effect estimate would represent a lower bound for the true effect. A similar argument holds for false negatives. If trash is harder to detect in dirtier environments (where the model may struggle to tell many different trash pieces apart), our corrected estimates would still be too conservative.

well-balanced across experimental groups. Only for two of our main variables, street type (based on OpenStreetMap) and the share of people engaging in voluntary work (based on survey data), we report significant differences between groups.

4 Results

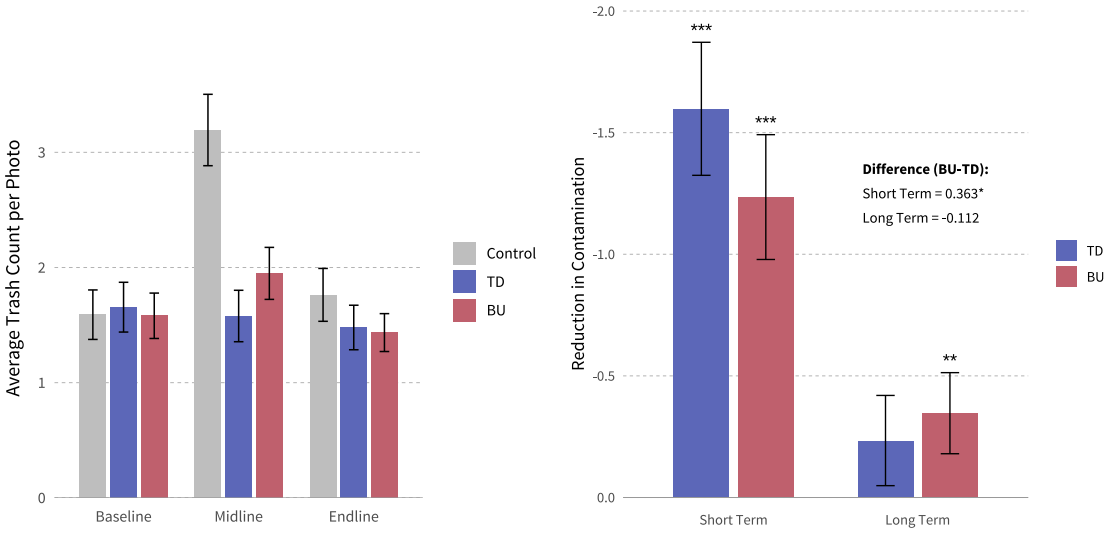
4.1 Contamination effects

To assess the causal effect of the two treatments on contamination levels for each post-treatment $wave \in \{midline, endline\}$, we use

$$Y_{iv}^{wave} = \beta_1 T_1 + \beta_2 T_2 + \delta Y_{iv}^{baseline} + \mu_s + \epsilon_{iv} \tag{2}$$

where Y_{iv}^{wave} is the midline or endline trash count for raster cell i in community v; T_1 and T_2 are treatment indicators for treatment 1 (top-down) and treatment 2 (bottom-up); $Y_{iv}^{baseline}$ is the baseline raster cell contamination level; and μ_s are strata fixed effects. Standard errors are clustered at the community level.

Figure 6: Contamination results



(a) Contamination over time

(b) Main contamination results

Notes: Both panels are based on trash count at the raster cell level. Panel (a) shows averages by treatment group and survey wave. Panel (b) shows treatment effects using our main specification (raster cells, controlling for baseline values and strata fixed effects). Error bars represent clustered standard errors at the community level. $^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

For our pre-registered **main specification** based on Equation 2, we find large *immediate effects* for both interventions (Figure 6 and Table 3, row 1). The top-

down intervention reduced solid waste contamination by 1.6 trash pieces or 50 percent ($p < 0.01$). The bottom-up intervention had a significantly smaller ($p < 0.1$), but still substantial impact of 1.24 trash pieces or 39 percent ($p < 0.01$). Estimates for *long-term impacts* reveal a stark depletion of effects relative to the control group after the withdrawal of external support (Table A7). Four months after the end of the treatment, communities in the top-down intervention outperform the control group by only 0.23 trash pieces or 14 percent, an effect that is statistically indistinguishable from zero at conventional levels ($p = 0.20$). This corresponds to a depreciation of about 72 percent compared to immediate effects. For the bottom-up intervention, we document a slightly larger and statistically significant long-term effect of 0.35 trash pieces or 21 percent ($p < 0.05$), corresponding to a depletion of 47 percent compared to immediate effects. The differences between treatments are not statistically significant for either long-term effects ($p = 0.47$) or depletion rates ($p = 0.25$). Importantly, this depletion largely reflects declining contamination in control communities rather than increasing contamination in treated communities (Figure 6). This general improvement may reflect seasonal variation, changes in municipal waste collection, or other external factors that we cannot directly identify.

To assess the **robustness** of our findings, we compare effects with alternative (i) model specifications and (ii) contamination measures. Results are very similar across different *model specifications*. They become slightly more pronounced and precisely estimated when adding controls or using the kernel approach for spatial aggregation (Table 3, rows 2–3 and 5–7), and slightly less so for raw community averages (Table A6). We can also compare our objective contamination results with effects based on *subjective contamination measures* (Table A5), namely manual photo assessments by three independent enumerators, on-site assessments by enumerators, and perceived community cleanliness from survey data (see Section 3.4.1). Effects are qualitatively very similar across most measures. Enumerator photo assessments exhibit the same pattern as our main results but are less precisely estimated, while on-site assessments show similar short-term results but less pronounced long-term effects. The strongest deviation from our main results appears in the survey measure, which shows smaller but highly persistent effects for both treatments, likely reflecting the absence of a specific time reference in the survey question (see Table A14). When including enumerator fixed effects, results become more pronounced and broadly consistent across all measures (except survey outcomes), with the largest adjustments to point estimates for on-site assessments

Table 3: Contamination results

	Short-term effects				*Long-term effects*			
	T1: TD	T2: BU	T2 - T1	N	T1: TD	T2: BU	T2 - T1	N
Raster approach								
No controls	-1.598***	-1.235***	0.363*	10260	-0.234	-0.347**	-0.112	10260
	(0.273)	(0.257)	(0.218)		(0.185)	(0.167)	(0.156)	
Minimal controls	-1.532***	-1.171***	0.362*	10260	-0.264	-0.319**	-0.055	10260
	(0.248)	(0.233)	(0.205)		(0.177)	(0.149)	(0.141)	
Full controls	-1.576***	-1.179***	0.397**	10260	-0.318*	-0.354**	-0.036	10260
	(0.238)	(0.222)	(0.198)		(0.180)	(0.157)	(0.140)	
Enumerator FEs	-1.704***	-1.445***	0.259	10260	-0.245	-0.358***	-0.114	10260
	(0.268)	(0.254)	(0.192)		(0.153)	(0.133)	(0.143)	
Kernel approach								
No controls	-1.567***	-1.143***	0.424**	60702	-0.170	-0.284**	-0.114	65671
	(0.252)	(0.249)	(0.210)		(0.161)	(0.144)	(0.149)	
Minimal controls	-1.491***	-1.084***	0.407**	60702	-0.203	-0.276**	-0.074	65671
	(0.230)	(0.229)	(0.202)		(0.154)	(0.130)	(0.134)	
Full controls	-1.543***	-1.106***	0.437**	60702	-0.255	-0.309**	-0.053	65671
	(0.227)	(0.224)	(0.199)		(0.159)	(0.140)	(0.131)	
Enumerator FEs	-1.608***	-1.260***	0.348*	60702	-0.202	-0.297***	-0.095	65671
	(0.242)	(0.241)	(0.188)		(0.134)	(0.115)	(0.132)	

All models control for baseline contamination and strata fixed effects. Minimal controls include raster/kernel-level street type and community remoteness (travel time to regional capital). Full controls additionally include community-level survey outcomes for community size, SES, social capital, and social norms at baseline. Standard errors are clustered at the community level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

and the smallest for photo-based measures.[12] This underscores the advantages of our objective deep learning-based measurement strategy.

We also assess the **heterogeneity of effects** along different dimensions, including street type, baseline contamination, community size, remoteness, social norms,

[12]While we did not pre-register enumerator fixed effects, there are compelling reasons to include them given that small groups of enumerators covered each community, which can introduce correlated measurement errors. Enumerator effects are particularly pronounced for on-site assessments and surveys, likely due to the greater scope for subjective judgment or influence (Table 1). While enumerator effects are also possible for photo-based measures, for example if enumerators systematically shift the camera to center on waste or vary how much of the street versus the background is captured, we observe very limited clustering by enumerator for these measures ($ICC = 0.009$ for our objective measure, see Table 1). Note that for enumerator photo assessments, each photo was assessed by the same three enumerators and the average was taken. Hence, enumerator fixed effects for this outcome refer to the person who took the photo rather than who assessed it.

social capital, and poverty, finding little indication of meaningful heterogeneous treatment effects (see Figures A4, A5, and A6). The one exception is that both interventions show significantly larger reductions in contamination on (larger) streets compared to smaller pathways (Figure A4). A likely explanation is that cleaning efforts were concentrated on (larger) streets.

4.2 Mechanisms

We distinguish between two broad pathways to bottom-up public goods provision (see Appendix Section A7 for a more detailed theoretical framework). A first channel operates through *individual behaviors and social norms*. Here, each person's contribution incrementally affects the overall level of provision. This mechanism corresponds to a continuous public goods problem, meaning that different levels of provision can be realized depending on the number of individual contributors. Examples include individual water conservation decisions, household sanitation practices, or forest harvesting behaviors. In this context, capacity-building can alleviate informational constraints that prevent communities from reaching higher levels of provision. This can happen by making people value the public good more or improving their knowledge on how to provide it, thereby directly increasing their willingness to contribute, or by correcting potential misconceptions regarding other people's willingness to contribute. In the context of our waste management intervention, such effects would be observable in people's waste disposal and littering behaviors and norms.

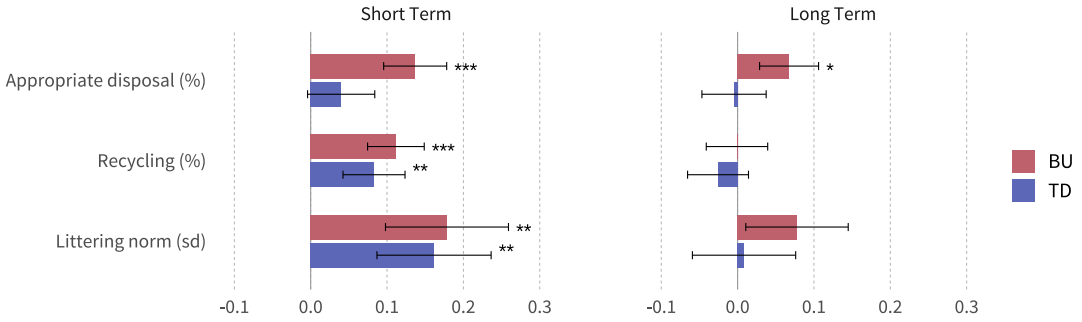
A second mechanism is related to *collective action and social capital*. Even when enough people are willing to contribute, public goods provision sometimes requires collective action and is thus only achieved if people manage to coordinate their efforts. Accordingly, the challenge related to this provision channel is better conceptualized as a threshold public goods problem, where provision either occurs or does not, depending on whether coordination succeeds. Examples include organizing irrigation systems, building a community road, or coordinating strikes, where individual contributions are necessary but not sufficient, and collective mobilization determines success. Yet, coordinating such actions entails transaction costs, with the magnitude of these costs depending on how well people know and trust each other, i.e., on the level of social capital. Here, capacity building can alleviate organizational constraints to public goods provision by building social capital or local leadership. For our waste management intervention, this collective action channel corresponds to organizing waste removal systems such as community cleanups.

While for some public goods only one of these channels is relevant (e.g., organizing a strike always requires collective action), for many others, both pathways are available. For instance, reducing carbon emissions can occur through individual consumption choices or through coordinated policy interventions. This is also true for local waste management, where provision can occur either through individual behavioral changes that prevent waste accumulation or through coordinated action to put in place waste collection systems. In this section, we explore what role these two mechanisms played in our study.

4.2.1 Waste-related behaviors and norm shifts

Based on our survey data, we find substantial short-term behavioral effects for the *bottom-up intervention* (see Figure 7 and Table 4; for variable definitions, see Table A14). The intervention increased the share of respondents reporting appropriate disposal practices by 14 percentage points ($p < 0.01$), driven by a reduction in burning (see Table A8), and increased recycling by 11 percentage points ($p < 0.01$). We also observe substantial effects of 0.18 standard deviations ($p < 0.05$) on the social norms index that appear to be due to shifts in descriptive norms (Appendix Table A8). The evidence for the persistence of these behavioral effects is more mixed. In the long term, disposal effects become markedly smaller but remain marginally significant (6.8 percentage points, $p = 0.08$), littering norms depreciate to 0.08 standard deviations ($p = 0.26$), and recycling effects disappear entirely.

Figure 7: Treatment effects on waste-related behaviors and social norms



Notes: Littering norms are an index of descriptive norms (share that litters), injunctive norms (share that disapproves of littering), and punishment norms (share that punishes littering). Appropriate disposal includes usage of truck or deposit rather than burning, burying, or dumping waste. For exact definitions, see Table A14. Regression specification is identical to Table 4. Error bars represent clustered standard errors at the community level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Interestingly, we observe very similar behavioral patterns for the *top-down inter-*

vention. We find substantial short-term effects on recycling (8.3 percentage points, $p < 0.05$), suggesting that residents made use of the collection service to separate and recycle their waste. Similarly, the top-down intervention induced substantial short-term (0.16 standard deviations, $p < 0.05$), but not long-term changes in littering norms. These behavioral effects for the top-down intervention align with an ample literature showing that people litter less in clean environments (Cialdini et al., 1990; Ramos and Torgler, 2012; Bateson et al., 2013; Sagebiel et al., 2020). A likely explanation is that residents form their beliefs about how many others are littering based on observed contamination levels and modify their behaviors based on this inferred social norm. This contrasts evidence from other settings showing that public provision can crowd out private contributions (Andreoni and Payne, 2003). While for recycling and littering norms top-down effects are statistically indistinguishable from bottom-up effects, household disposal practices constitute a notable exception to this pattern. Here the effects we observe for the bottom-up treatment are absent in the top-down intervention, pointing to a potential additional benefit beyond what is reflected in street contamination, as reductions in waste burning represent an environmental improvement not captured by our main outcome measure.

Overall, while we find substantial short-term improvements in behavioral littering responses, most of these gains disappear after the withdrawal of support and are similar in magnitude to those observed under the top-down intervention. These patterns suggest that shifts in waste-related behaviors and social norms may have played a limited role in bottom-up public goods provision in our setting. This is consistent with a growing body of evidence showing that achieving and sustaining high levels of public goods provision through individual behavioral change alone can be challenging (e.g., Kaiser et al., 2025; Fischbacher and Gächter, 2010; Fischbacher et al., 2001). The behavioral adjustments in response to observed contamination further suggests that maintaining high levels of cleanliness through behavioral changes alone would require near-full compliance with non-littering norms. In the absence of collection systems, waste generated by non-compliers accumulates, discouraging others from maintaining proper disposal practices, and ultimately pushing the community back toward low levels of provision.

4.2.2 Social capital and collective action

Looking at *survey outcomes* related to social capital, we find large and persistent effects for the bottom-up treatment and no effects for the top-down intervention (see Figure 8 and Table 4; for variable definitions, see Table A14). The composite

Table 4: Survey regression results

	Short-term effects			*Long-term effects*		
	T1: TD	T2: BU	T2 - T1	T1: TD	T2: BU	T2 - T1
Waste-related behaviors						
Appropriate disposal (%)	0.040	0.137***	0.097**	-0.005	0.067*	0.072*
	(0.044)	(0.041)	(0.046)	(0.042)	(0.039)	(0.040)
Recycling (%)	0.083**	0.112***	0.029	-0.026	-0.001	0.025
	(0.041)	(0.037)	(0.029)	(0.040)	(0.040)	(0.040)
Recycling items (count)	0.092	0.267**	0.175	-0.162	0.055	0.216**
	(0.113)	(0.125)	(0.111)	(0.098)	(0.102)	(0.095)
Littering norms						
Littering norms index (sd)	0.162**	0.178**	0.017	0.008	0.078	0.069
	(0.075)	(0.081)	(0.079)	(0.068)	(0.067)	(0.074)
Descriptive norm (sd)	0.244***	0.308***	0.064	0.002	0.066	0.064
	(0.089)	(0.091)	(0.084)	(0.069)	(0.066)	(0.072)
Injunctive norm (sd)	0.042	0.009	-0.033	-0.020	0.039	0.058
	(0.073)	(0.078)	(0.075)	(0.083)	(0.073)	(0.080)
Punishment (sd)	-0.002	0.000	0.002	0.039	0.039	0.000
	(0.057)	(0.058)	(0.064)	(0.074)	(0.073)	(0.074)
Social capital						
Social capital index (sd)	0.126	0.336***	0.211**	0.006	0.183**	0.177**
	(0.084)	(0.087)	(0.095)	(0.078)	(0.083)	(0.084)
Networks (sd)	0.163	0.180	0.017	0.087	0.079	-0.008
	(0.127)	(0.110)	(0.123)	(0.096)	(0.089)	(0.095)
Trust (sd)	0.138	0.119	-0.019	-0.091	0.019	0.110
	(0.111)	(0.114)	(0.119)	(0.095)	(0.096)	(0.083)
Organizations (%)	-0.007	0.042	0.049*	-0.009	0.011	0.020
	(0.023)	(0.026)	(0.028)	(0.023)	(0.025)	(0.027)
Volunteering (%)	0.011	0.159***	0.149***	0.035	0.129***	0.094***
	(0.031)	(0.034)	(0.037)	(0.029)	(0.031)	(0.031)

Sample sizes are n = 2066 for the estimation of short-term effects and n = 1832 for long-term effects. Controls include strata fixed effects, sex, age, education (dummies), and the baseline value for the respective outcome. Standard errors are clustered at the community level. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$

social capital index increases by 0.33 standard deviations in the short run ($p < 0.01$) and 0.18 standard deviations in the long run ($p < 0.05$). All social capital subcomponents have positive coefficients, with marginally significant short-term effects for networks, and substantial and highly significant effects for volunteering of 16 percentage points in the short run ($p < 0.01$) and 13 percentage points in the long run ($p < 0.01$). An important question is whether these effects correspond to a general increase in volunteering and thus social capital that might help the community address public goods challenges beyond waste management more effectively. When examining participation in community cleanups specifically, effects are even larger, namely 30 percentage points in the short run and 17 percentage points in the long run (Table A10).[13] This makes it likely that the volunteering increase was largely driven by waste-related activities rather than broader forms of community engagement.[14]

The patterns we observe in our survey data are closely mirrored in our *activity registry*, where we observe an even starker decline in collective action in the post-intervention period (Figure 9). During the intervention period, roughly 90 percent of communities conducted at least one cleanup per month. This share immediately drops to less than half after the facilitator withdrawal and declines further to roughly 25 percent ten months later.[15] Notably, conditional on campaigns taking place, participation remained stable at around 20 participants per cleanup. This suggests that the key challenge to sustained collective action was not declining willingness to participate, but rather the difficulty of organizing campaigns in the first place.

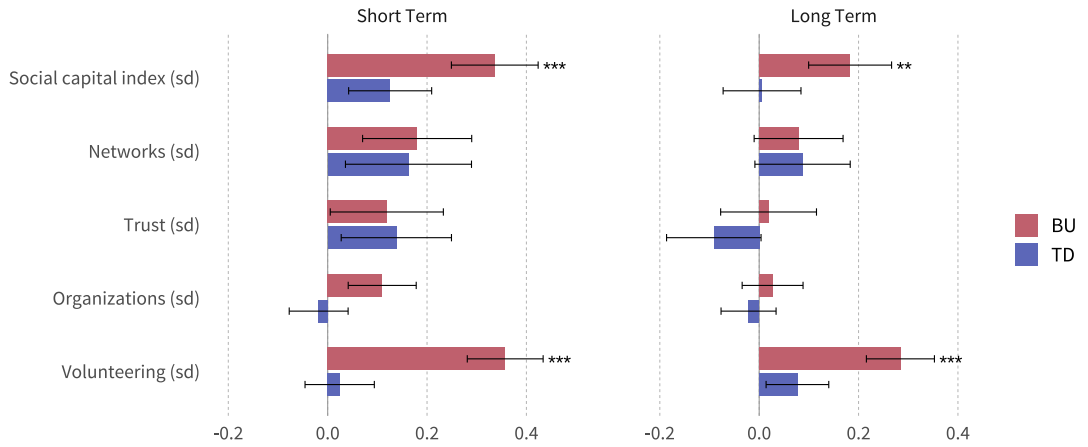
Overall, these patterns point to collective cleaning as the *primary mechanism*

[13]These participation effects appear implausibly high given community size and actual cleanup participation. With an average community size of 305, a 30 percentage point increase would imply 92 additional cleaning volunteers per community. Yet our registry data documents an average of 18.7 participants per cleanup across 3.4 cleanups during the intervention period. Even assuming no overlap between participants across cleanups, this yields only 64 unique volunteers per community, which is well below the 92 implied by the survey data. Likely explanations include selection into the survey sample, where respondents may be more likely than the average resident to participate in community activities (e.g., because they spend more time in the community or are more pro-social, affecting their propensity to participate in both the survey and cleanups), social desirability bias amplified by the intervention, or people implicitly reporting volunteering at the household level.

[14]Note that due to the binary nature of the volunteering and cleanup variables, we cannot fully disentangle whether the treatment affected volunteering only through increased cleanup participation or also through independent effects. Even if volunteering were merely a noisy representation of cleanup participation, conditioning on cleanup participation would not eliminate treatment effects, as discretization prevents regression from correctly separating the effects of two correlated variables.

[15]Note that as we cannot observe this outcome for the control group, we do not know if this reflects an enduring effect on collective action or a reversion to the control group mean.

Figure 8: Treatment effects on social capital

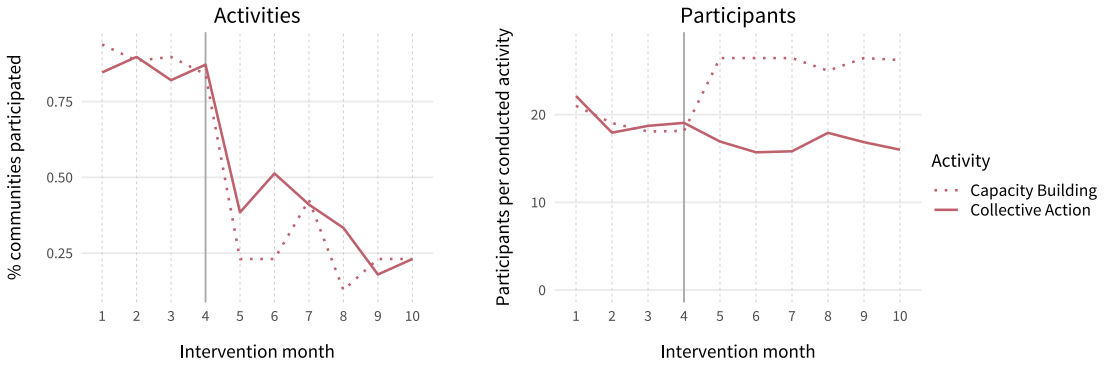


Notes: Social capital is an index of networks (number of people known), trust, organizations (membership in community groups), and volunteering. For exact definitions, see Table A14. Regression specification is identical to Table 4. Error bars represent clustered standard errors at the community level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

through which the bottom-up treatment reduced contamination. The high participation rate in the first treatment month indicates that even without intense prior capacity-building and awareness activities, a sufficiently high share of residents were willing to contribute to a clean community. The low levels of provision we observe absent any treatment are thus likely the result of coordination failure rather than insufficient willingness to contribute. Accordingly, the high levels of collective action and volunteering during the treatment period suggest that the bottom-up intervention was successful at overcoming this coordination failure by alleviating organizational constraints and bringing residents together for effective bottom-up provision. However, the large decline in collective action after the withdrawal of support indicates that much of the success along this organizational dimension was tied to the presence of the facilitator. Mobilizing for collective action is time-consuming and demands a disproportionate contribution from the person (or persons) taking the lead. If people are only willing to contribute about as much as others do, no such leader will emerge to take on this burden. During the intervention period, this leadership problem was temporarily solved through the leadership of the facilitator, but in many communities, the intervention failed to establish enduring structures of local organization and leadership. This interpretation is reinforced by qualitative evidence from interviews with community members who assumed responsibility after the facilitator's departure, where "time constraints to mobilize people and organize campaigns" emerged as the most frequently mentioned challenge to project

continuation.

Figure 9: Collective action in bottom-up communities over time



Notes: Capacity-building activities include educational sessions, practical workshops, awareness campaigns, and meetings. Collective action activities refer to community cleanups. Months 1-4 correspond to the intervention period.

4.3 Cost-effectiveness

We also collected detailed cost data for each treatment. This allows us to compare the cost-effectiveness of each delivery model (i) in the intervention period and (ii) in the longer run. The cost data can be combined with insights from our registry and survey to better understand (iii) the efficiency of waste collection efforts under each model, (iv) the financial sustainability of a paid service delivery model, and (v) who bore the costs of bottom-up provision.

We begin by assessing the **cost-effectiveness in the intervention period**. The top-down intervention had an average monthly operational cost of 98 USD per community (see Figure 10a), with main cost items being waste collector salaries and transportation costs (gasoline and truck depreciation). This corresponds to a cost-effectiveness of approximately 1.6 fewer trash pieces in the average photo location per 100 USD invested. With an average monthly cost of 216 USD per community, the bottom-up treatment cost more than twice as much, mainly due to expenditures for snacks during community activities and facilitator salaries. This yields an effect of 0.6 trash pieces per 100 USD, suggesting that in the short term, direct public good provision was three times more cost-effective than investing in capacity building for bottom-up provision. If we additionally account for people's volunteering time in the cleanup campaigns, the cost further increases to 375 USD, and cost-effectiveness decreases to 0.3 trash pieces per 100 USD, making top-down provision five times more cost-effective.[16]

[16]Estimates are based on the number of participants and the duration of all cleaning campaigns

Figure 10: Costs and cleanup statistics by intervention



(a) Intervention costs

(b) Cleanup statistics

Notes: Panel (a) shows average monthly operational costs per community. Volunteering time is valued at the local minimum wage; volunteering costs overlap with food and catering costs as we consider snacks as a form of compensation for volunteering. Panel (b) shows community averages per cleanup event from the registry data.

The key argument for community-based initiatives is that initial investments into local capacity pay off through sustainable improvements without further external support. We pursue two strategies to compare **long-term cost-effectiveness** between interventions. First, we relate post-intervention effects to average monthly expenditure during the intervention period. This yields 0.24 trash pieces on an average photo per 100 USD for the top-down treatment and 0.16 (or 0.09 accounting for volunteering time) for the bottom-up treatment, suggesting that four months after withdrawal, the higher initial investment for bottom-up provision has not paid off. Second, we consider what sustained provision would require for each model. For top-down provision, continuation would likely look identical to the intervention period, yielding the same cost-effectiveness of 1.6 trash pieces per 100 USD. For

recorded in the registry. We value people's time at the local minimum wage of 365 USD, which also corresponds to the wage paid to the trash collectors in the top-down intervention. We convert this into an hourly wage using the average number of monthly working days and daily working hours of the top-down cleaning team. This yields a total cost of 159 USD. We further consider the snacks as a form of payment, which is why the hatched bar for "volunteering" overlaps with the gray bar for "food and catering". Note that the actual opportunity cost of volunteer time may be lower than the minimum wage for individuals who are unemployed or underemployed, or who derive utility from participation itself. While we cannot directly assess if this is the case, our data are consistent with this possibility: 92 percent of survey respondents state that they would be willing to contribute time to a community project, but only 39 percent state that they would contribute money, and this gap widens further for poorer communities.

bottom-up provision, in the absence of radical and sustained shifts in social norms (see Section 4.2), communities would need to continue organizing regular cleanup events. This means that at a minimum, they would have to incur volunteering costs of 159 USD per monthly cleanup, resulting in a cost-effectiveness of 0.8 trash pieces per 100 USD. While such a system of sustained collective cleanups failed to materialize in most communities, this calculation shows that even if it were to succeed, it would not outperform top-down provision when volunteer time is accounted for.

A related question for long-term cost-effectiveness is whether service delivery crowds out private provision – a common criticism of top-down models (Andreoni and Payne, 2003). Our results suggest the contrary. For our top-down intervention, consistent with the observed positive short-term effects on littering norms and participation in cleanups (see Tables 4 and A10), we also find no long-term reduction in willingness to pay for cleaning after the removal of support. This suggests that, if anything, direct service delivery reinforced collaborative behavior rather than discouraging it, implying cost reductions rather than increases over time.

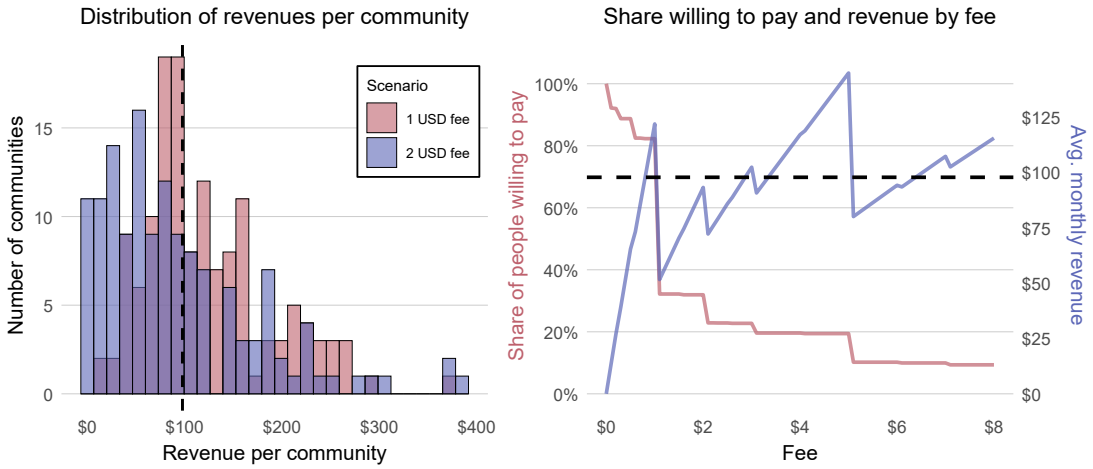
To better understand the **efficiency of waste collection** in each delivery model, we compare the cleanup logs between both interventions. In the top-down treatment, the average cleanup involved 3.6 people and lasted 2.2 hours, resulting in 8 worker-hours. In contrast, in the bottom-up treatment, 19.4 volunteers participated in the average cleanup and, at roughly 2.4 hours, it took them even slightly longer, resulting in a total of 50 worker-hours. Plausible explanations are that in bottom-up provision, benefits of specialization cannot be realized or that coordinating the effort in a large group may be inefficient and result in most people not contributing meaningfully most of the time.

Our data also allows us to assess whether it would be financially sustainable to offer **waste collection as a paid service** to residents. To this end, we collected data on respondents' willingness to pay (WTP) for the service offered in the top-down treatment as well as their beliefs about the average community member's respective WTP. The two distributions are remarkably similar, with winsorized means of 2.28 USD and 2.27 USD respectively (median of 1 USD for both), suggesting minimal social desirability bias in individual reports. Assuming survey participants' reported WTP distributions are representative for the working-age adult population in each community, we compute potential revenues under different fee structures.[17] The highest total revenues are raised with a 5 USD fee (19% con-

[17]We winsorize WTP at the 95th percentile (10 USD for own WTP, 9 USD for perceived WTP of others) to address extreme outliers. We define the working-age adult population as individuals aged 20-64 and use the population share for Morazán (55%) to estimate this figure based on each community's total population. Even though the question was framed individually, respondents

tribute, 145 USD raised per community) and with a 1 USD fee (82% contribute, 122 USD per community), and in both cases, the total revenue exceeds the average operational costs of 98 USD.[18] These calculations suggest that a region-wide voluntary paid service model might be profitable, although its viability rests on several strong assumptions that we cannot test directly. The substantial WTP further reinforces the conclusion that many residents are willing to contribute to a clean environment but communities fail to coordinate to achieve effective provision (Section 4.2).

Figure 11: Willingness to pay for waste collection



Notes: Left panel shows the distribution of total revenue per community under 1 USD and 2 USD monthly fees. Right panel shows the share willing to pay (left axis) and average monthly revenue (right axis) by fee level. Revenue calculations assume survey WTP is representative of working-age adults (55% of population). WTP is winsorized at the 95th percentile. Dashed line indicates average operational costs (98 USD).

Finally, we can also analyze **who bore the costs of bottom-up provision** (Table A12). Our activity registry shows that 85 percent of volunteers in cleanups were women (52%) or children (33%). Participation rates also tend to be slightly

might have interpreted it at the household level. We thus also perform calculation based on WTP as household-level contributions, using average household size of 3.1 people. Under this scenario, the total revenue would be considerably lower and not reach financial sustainability under any fee (see Figure A7).

[18]Note that the 1 USD scenario is more likely to represent a stable equilibrium, as its broad support may prevent residents from stopping to contribute upon learning about defectors – though this is an assumption we cannot test with our data. While average revenue under a 1 USD fee exceeds operational costs (98 USD), contributions vary substantially across communities, and only 55 percent raise sufficient revenue to cover average costs. This suggests that the overall profitability of the service is driven by a minority of high-revenue communities, although costs could also vary with community size, remoteness, and contamination levels, which are not accounted for in our revenue calculations. Note further that costs might decrease if the program were scaled, as transportation expenses could fall relative to our experimental setup, which required communities to be spatially separated.

higher in poorer communities (20 vs. 17 participants per cleanup). Our survey data further shows that people in poorer communities report higher volunteering levels and willingness to contribute time to community projects. At the same time, poorer communities are substantially less likely to be served by municipal garbage trucks (see Table A12). This is consistent with the notion that these communities compensate for lower levels of top-down service provision by the government through bottom-up collective action. Overall, these results point to an important equity concern for bottom-up initiatives. If not carefully designed, such programs may act as a form of regressive taxation and place additional burdens on poor communities or marginalized groups.

5 Conclusion

While the standard solution to the public goods problem calls for a top-down intervention by a powerful actor such as the state, vast empirical evidence shows that groups sometimes manage to coordinate to provide or protect local public goods bottom up. This has led to the view that a one-time investment into local capacity may enable communities to guarantee effective and sustainable bottom-up provision without continued external support. This perspective has been particularly influential in development cooperation, contributing to a strategic shift from serving poor communities to empowering them to address local challenges through their own efforts. Despite their increasing popularity over the last few decades, it remains largely unclear how such interventions compare to the direct top-down provision schemes they replace. In this study, we present the results of a randomized controlled trial comparing the effectiveness of bottom-up and top-down strategies to address local waste pollution in rural El Salvador.

We find that bottom-up provision was effective at reducing local waste pollution, with immediate reductions of roughly 40 percent relative to the control group. Effects depreciate sharply after the removal of external support, but remain statistically significant at about 21 percent four months later. Beyond contamination, we also document effects on waste-related behaviors and social norms, as well as substantial and sustained effects on social capital, which are driven by a large increase in volunteering. Yet, bottom-up provision was not more effective than the top-down alternative. In the short run, effects are substantially smaller than under top-down service delivery, where we estimate reductions of roughly 50 percent. The comparatively lower effects in the bottom-up treatment are achieved at more than twice the monetary costs and almost four times the total costs when volunteer time

is included. Organizing bottom-up provision also proved substantially less efficient. It required roughly six times as many worker-hours per average cleanup, implying substantial time costs for participants, most of whom were women and children. We find only weak evidence for higher sustainability, as long-term effects under top-down provision are only slightly and statistically insignificantly smaller, and social-norm effects are largely similar across interventions. While social capital effects are confined to the bottom-up treatment, they appear to be largely driven by participation in cleanup campaigns. Overall, these patterns are consistent with a theoretical framework in which individuals are willing to contribute (conditionally) to a clean community, but fail to coordinate effective provision in the absence of a committed leader. During the intervention, this collective action problem was temporarily solved through the leadership of the facilitator, but in most communities no enduring organizational structures or local leadership persisted after the withdrawal of support.

Our findings have several important implications for our understanding of local public goods provision and for the design of interventions to address underprovision. First, they suggest that directly providing public goods – or strengthening state capacity to do so – may outperform investments in parallel bottom-up structures. While groups sometimes manage to coordinate bottom-up provision organically, promoting such coordination through external capacity-building interventions may be challenging. Our results show that inducing bottom-up provision can be effective, but that the costs of coordinating collective action are high and that sustainability through one-time "seed" investments is difficult to achieve. Second, our study highlights an important dimension of participatory development that has received little attention in prior research and policy debates. Bottom-up programs do not only shift power, but also responsibilities, potentially placing additional burdens on poor communities or marginalized groups. When designing participatory interventions, it is therefore crucial to consider the extent to which they genuinely benefit and empower the people they are intended to serve.

Despite these cautionary notes, our findings also highlight the promise of bottom-up approaches. We document substantial and broad effects spanning contamination, social norms, household disposal practices, and social capital, indicating that participatory strategies can successfully mobilize communities. This indicates that combined approaches, where top-down delivery is complemented by participatory bottom-up elements, may be a promising avenue for future research. Relatedly, more work is also needed to understand whether our findings generalize to other contexts, public goods, and intervention types.

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A Appendix

A1 Expert prediction survey results

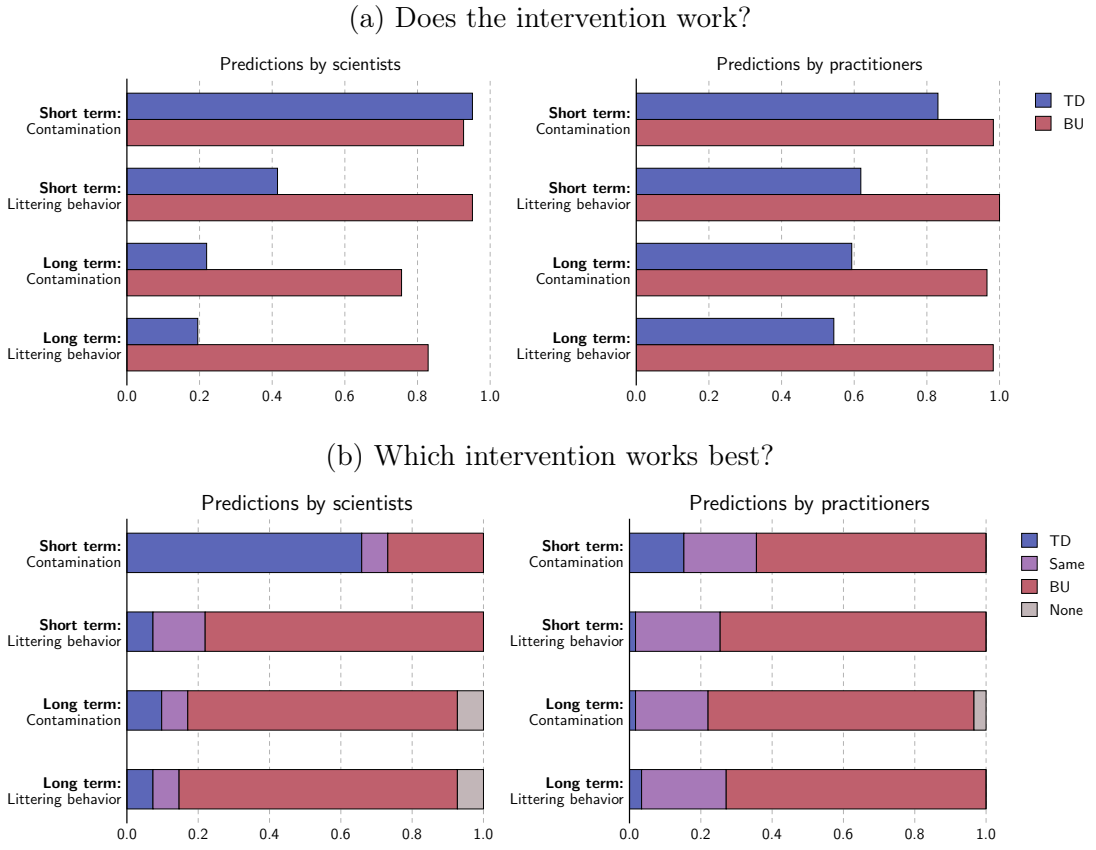
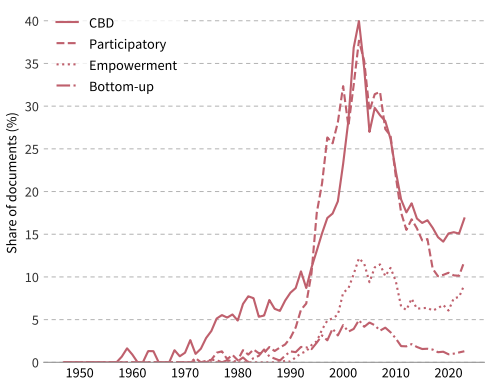


Figure A1: Pre-Survey Results

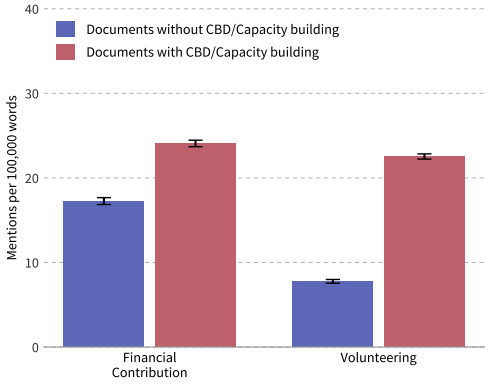
Notes: Illustration based on a prediction survey with 41 social scientists and 59 local experts. Upper panel shows the percentage of respondents who expect each intervention to have a positive effect. Lower panel shows the share of respondents indicating that a particular intervention worked best.

A2 World Bank documents analysis

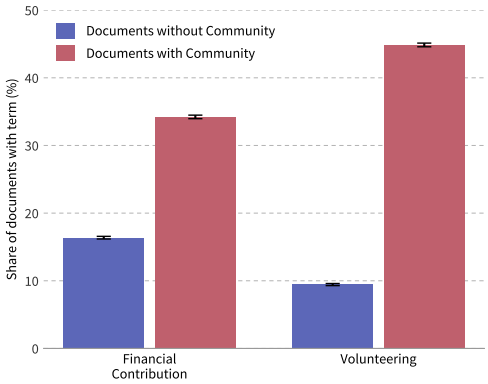
Figure A2: Terminology trends and contribution keywords by approach



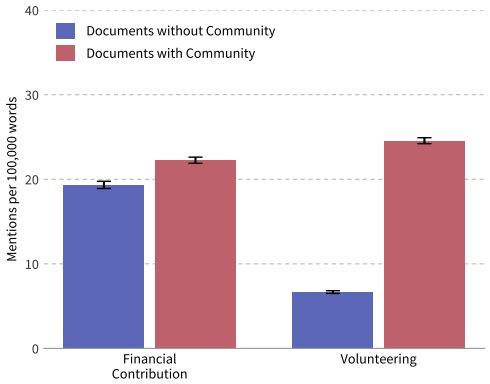
(a) CBD terminology trends



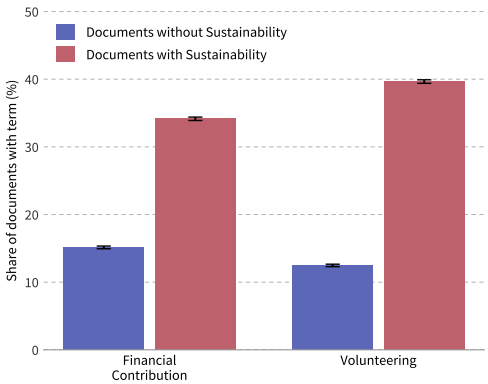
(b) CBD/CB: Word frequencies



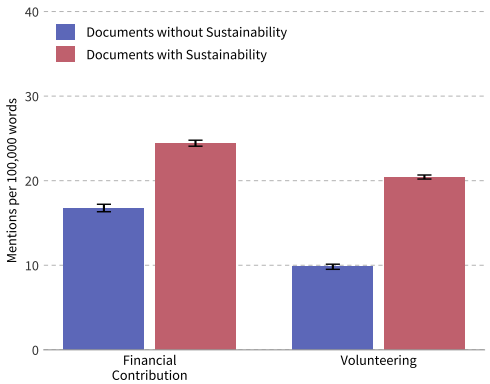
(c) Community: Document shares



(d) Community: Word frequencies



(e) Sustainability: Document shares



(f) Sustainability: Word frequencies

Notes: Illustration based on World Bank project documents (see Figure 1 for details). Panel (a) shows CBD terminology trends. Panels (b)–(f) show the frequency of contribution-related words (either as document shares or per 100,000 words), comparing documents mentioning CBD/capacity building, community, or sustainability (red) with those not mentioning these terms (blue).

A3 Baseline characteristics and attrition

Table A1: Balance at baseline

	Control	T1: TD	T2: BU	P-value	N
Photo trash detection					
Raster approach (count)	0.575	0.622	0.587	0.935	10260
Kernel approach (count)	0.569	0.612	0.579	0.963	60702
Raw averages (count)	0.593	0.590	0.597	0.968	120
Enumerator photo assessment					
Raster approach (1-4)	1.567	1.592	1.576	0.606	1897
Kernel approach (1-4)	1.580	1.579	1.579	0.931	6173
Raw averages (1-4)	1.578	1.578	1.579	0.962	120
Enumerator onsite assessment					
Raster approach (1-4)	1.996	1.972	1.979	0.291	4074
Kernel approach (1-4)	1.991	1.982	1.976	0.238	12211
Raw averages (1-4)	2.011	1.962	2.021	0.394	120
Geographic characteristics					
Street type: Major road (%)	0.096	0.066	0.079	0.749	10260
Street type: Residential (%)	0.071	0.226	0.105	0.017	10260
Street type: Connector (%)	0.206	0.092	0.065	0.022	10260
Street type: Other (%)	0.431	0.447	0.559	0.201	10260
Street type: None (%)	0.196	0.169	0.192	0.762	10260
Travel time to regional capital (min)	25.751	24.856	26.766	0.793	10260
Survey: Sociodemographics					
Community size	305.349	300.439	294.964	0.785	2421
Poverty (1-5)	2.838	2.954	3.021	0.169	2421
Female	0.740	0.756	0.724	0.446	2421
Age	42.881	42.186	43.417	0.435	2421
Highest degree: None (%)	0.185	0.206	0.210	0.468	2421
Highest degree: Incomplete primary (%)	0.451	0.435	0.466	0.441	2421
Highest degree: Complete primary (%)	0.113	0.122	0.104	0.661	2421
Highest degree: High school (%)	0.201	0.185	0.186	0.696	2421
Highest degree: Tertiary (%)	0.051	0.052	0.035	0.280	2421
Survey: Contamination and waste disposal					
Perceived cleanliness (1-5)	3.146	3.103	3.118	0.587	2421
Appropriate disposal (%)	0.452	0.532	0.489	0.530	2421
WTP for cleaning (log)	0.352	0.374	0.315	0.888	2421
Survey: Social norms					
Littering norms index (sd)	0.000	-0.132	-0.063	0.120	2421
Descriptive norm (sd)	0.000	-0.052	0.060	0.183	2421
Injunctive norm (sd)	0.000	-0.067	0.031	0.619	2421
Punishment (sd)	0.000	-0.101	-0.195	0.002	2421
Survey: Social capital					
Social capital index (sd)	0.000	0.067	0.047	0.833	2421
Networks (sd)	0.000	0.125	-0.120	0.227	2421
Trust (sd)	0.000	-0.119	-0.062	0.423	2421
Organizations (%)	0.174	0.191	0.207	0.445	2421
Volunteering (%)	0.252	0.296	0.338	0.036	2421

We report the p-value of a joint F-test that each treatment dummy coefficient is equal to 0. Street type variables refer to the share of raster cells intersecting with each street type. Standard errors are clustered at the community level.

Table A2: Attriter characteristics at midline by treatment group

	Control	T1: TD	T2: BU	P-value	N
Sociodemographics					
Female	0.706	0.664	0.670	0.936	355
Age	47.360	40.899	41.810	0.002	355
Education	2.147	2.361	2.360	0.464	355
Poverty (1-5)	2.993	2.898	2.991	0.791	355
Community size	305.463	279.706	269.530	0.645	355
Contamination and waste disposal					
Perceived cleanliness (1-5)	3.154	3.008	3.290	0.102	355
Appropriate disposal (%)	0.353	0.445	0.460	0.164	355
WTP for cleaning (log)	0.478	0.479	0.150	0.122	355
Social norms					
Littering norms index (sd)	0.126	-0.045	-0.080	0.352	355
Descriptive norm (sd)	0.037	-0.083	-0.006	0.686	355
Injunctive norm (sd)	0.058	-0.025	0.035	0.934	355
Punishment (sd)	0.114	0.035	-0.162	0.249	355
Social capital					
Social capital index (sd)	-0.238	-0.032	-0.139	0.184	355
Networks (sd)	-0.056	0.277	-0.108	0.068	355
Trust (sd)	-0.089	-0.184	-0.149	0.809	355
Organizations (%)	0.110	0.109	0.110	0.911	355
Volunteering (%)	0.154	0.255	0.300	0.070	355

The first three columns represent attriter group means for each treatment. The last column indicates the p-value of a joint F-test that each treatment dummy coefficient is equal to 0. Education refers to highest completed degree: None = 1, incomplete primary = 2, complete primary = 3, high school degree = 4), technical = 5, and university degree = 6. Standard errors are clustered at the community level.

Table A3: Attriter characteristics at endline by treatment group

	Control	T1: TD	T2: BU	P-value	N
Sociodemographics					
Female	0.660	0.724	0.616	0.088	589
Age	40.994	39.654	41.421	0.442	589
Education	2.546	2.599	2.522	0.687	589
Poverty (1-5)	2.756	2.884	2.967	0.493	589
Community size	300.289	288.507	282.854	0.620	589
Contamination and waste disposal					
Perceived cleanliness (1-5)	3.093	3.115	3.129	0.999	589
Appropriate disposal (%)	0.330	0.498	0.478	0.112	589
WTP for cleaning (log)	0.473	0.458	0.463	0.992	589
Social norms					
Littering norms index (sd)	-0.018	-0.110	-0.008	0.523	589
Descriptive norm (sd)	-0.054	-0.094	0.155	0.027	589
Injunctive norm (sd)	0.012	-0.037	-0.003	0.880	589
Punishment (sd)	0.013	-0.051	-0.166	0.206	589
Social capital					
Social capital index (sd)	-0.123	-0.134	-0.093	0.909	589
Networks (sd)	-0.007	0.131	-0.138	0.441	589
Trust (sd)	-0.077	-0.222	-0.117	0.435	589
Organizations (%)	0.144	0.143	0.135	0.923	589
Volunteering (%)	0.201	0.195	0.317	0.061	589

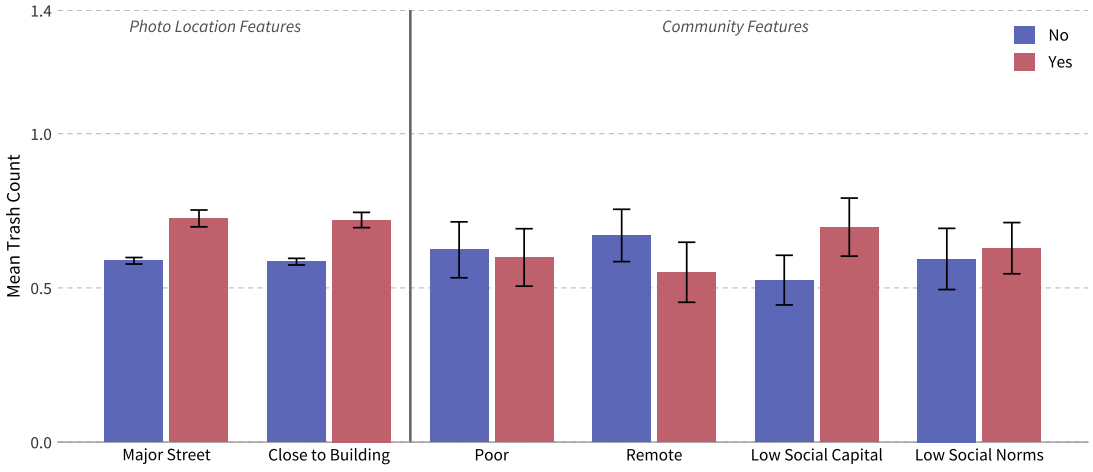
The first three columns represent attriter group means for each treatment. The last column indicates the p-value of a joint F-test that each treatment dummy coefficient is equal to 0. Education refers to highest completed degree: None = 1, incomplete primary = 2, complete primary = 3, high school degree = 4), technical = 5, and university degree = 6. Standard errors are clustered at the community level.

Table A4: Attrition by treatment group

	Control	T1: TD	T2: BU	P-value	N
Attrition midline	0.160	0.151	0.128	0.423	2421
Attrition endline	0.228	0.275	0.228	0.131	2421

The last row indicates the p-value of a joint F-test that each treatment dummy coefficient is equal to 0. Standard errors are clustered at the community level.

Figure A3: Contamination descriptives



Notes: Mean trash count per photo at baseline by location and community characteristics. Photo location features use photo-level data with simple standard errors. Community features (created by median splits) use clustered standard errors at the community level. Error bars represent 95% confidence intervals.

A4 Additional contamination results

Table A5: Objective vs. subjective contamination measures (standardized)

	Short-term effects				*Long-term effects*			
	T1: TD	T2: BU	T2 - T1	N	T1: TD	T2: BU	T2 - T1	N
Objective: Photo trash detection (std)								
No controls	-0.805***	-0.622***	0.183*	10260	-0.159	-0.236**	-0.077	10260
	(0.138)	(0.129)	(0.110)		(0.126)	(0.113)	(0.106)	
Enumerator FEs	-0.859***	-0.728***	0.130	10260	-0.166	-0.244***	-0.077	10260
	(0.135)	(0.128)	(0.097)		(0.104)	(0.091)	(0.098)	
Subjective: Enumerator photo assessment (std)								
No controls	-0.671***	-0.446***	0.225*	1897	-0.140	-0.207	-0.066	1897
	(0.154)	(0.142)	(0.136)		(0.169)	(0.141)	(0.169)	
Enumerator FEs	-0.722***	-0.517***	0.205	1897	-0.272*	-0.376***	-0.104	1897
	(0.156)	(0.127)	(0.146)		(0.143)	(0.140)	(0.160)	
Subjective: Enumerator on-site assessment (std)								
No controls	-0.878***	-0.697***	0.181	4074	-0.026	-0.038	-0.012	4074
	(0.178)	(0.168)	(0.186)		(0.217)	(0.175)	(0.236)	
Enumerator FEs	-1.045***	-0.788***	0.257	4074	-0.323**	-0.361***	-0.038	4074
	(0.150)	(0.160)	(0.165)		(0.139)	(0.130)	(0.159)	
Subjective: Perceived contamination from survey (std)								
No controls	-0.345*	-0.356*	-0.011	2066	-0.372**	-0.393**	-0.021	1832
	(0.193)	(0.189)	(0.174)		(0.175)	(0.182)	(0.174)	
Enumerator FEs	-0.287**	-0.364**	-0.078	2039	-0.374**	-0.343**	0.032	1822
	(0.131)	(0.142)	(0.131)		(0.172)	(0.173)	(0.173)	

Results reported in standard deviations at the community level. Perceived contamination from survey is reverse-coded so that higher values indicate more contamination. All models control for baseline contamination and strata fixed effects. Enumerator FE models additionally control for enumerator fixed effects. Standard errors are clustered at the community level. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$

Table A6: Photo trash detection using community averages

	Short-term effects				*Long-term effects*			
	T1: TD	T2: BU	T2 - T1	N	T1: TD	T2: BU	T2 - T1	N
No controls	-1.608***	-1.210***	0.398*	120	-0.213	-0.316*	-0.104	120
	(0.218)	(0.219)	(0.221)		(0.164)	(0.165)	(0.166)	
Minimal controls	-1.721***	-1.253***	0.468**	120	-0.333**	-0.401**	-0.068	120
	(0.225)	(0.226)	(0.220)		(0.164)	(0.164)	(0.160)	
Full controls	-1.791***	-1.281***	0.509**	120	-0.385**	-0.435***	-0.051	120
	(0.229)	(0.227)	(0.221)		(0.165)	(0.164)	(0.160)	

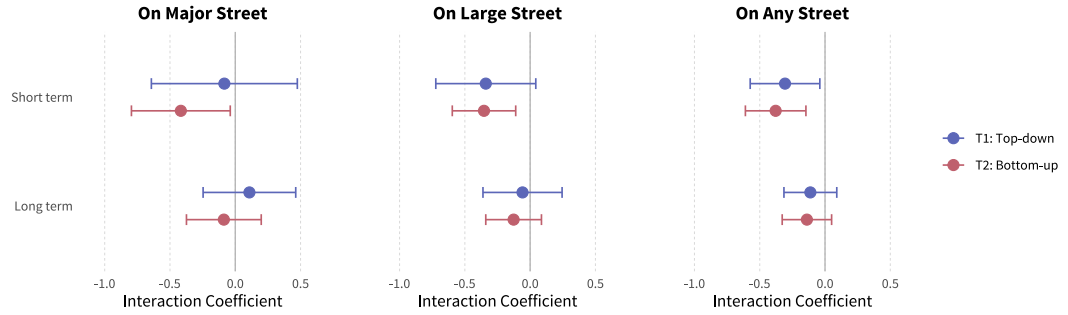
All models control for baseline contamination and strata fixed effects. Minimal controls include community-level street type and community remoteness (travel time to regional capital). Full controls additionally include community-level survey outcomes for community size, SES, social capital, and social norms at baseline. No clustering of standard errors as observations are at the community level. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Table A7: Treatment effect depletion rates

	T1: TD			T2: BU			Difference
	Short term	Long term	Depletion	Short term	Long term	Depletion	
Absolute	-1.598***	-0.234	1.364***	-1.235***	-0.347**	0.889***	0.475***
Relative	-0.504***	-0.139	0.724***	-0.389***	-0.206**	0.471**	0.253

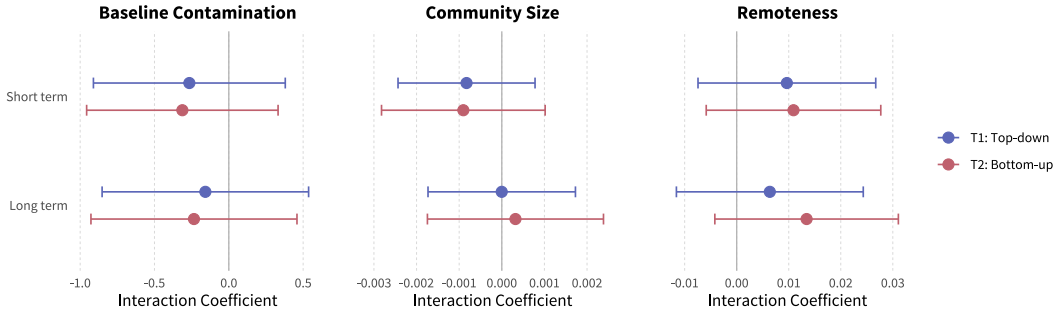
Estimates are based on the main specification (raster approach controlling for strata fixed effects and baseline values). Absolute effects are measured in trash count. Relative effects are expressed as a proportion of the control group mean. Depletion rate shows the change from short-term to long-term effects (absolute: total effect lost; relative: fraction of the relative effect lost). Control group means (bias-corrected) are 3.17 at short-term and 1.68 at long-term. Absolute depletion estimates use clustered standard errors at the community level; relative depletion estimates use cluster bootstrap standard errors (1,000 replications). $*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Figure A4: Effect heterogeneity by street type



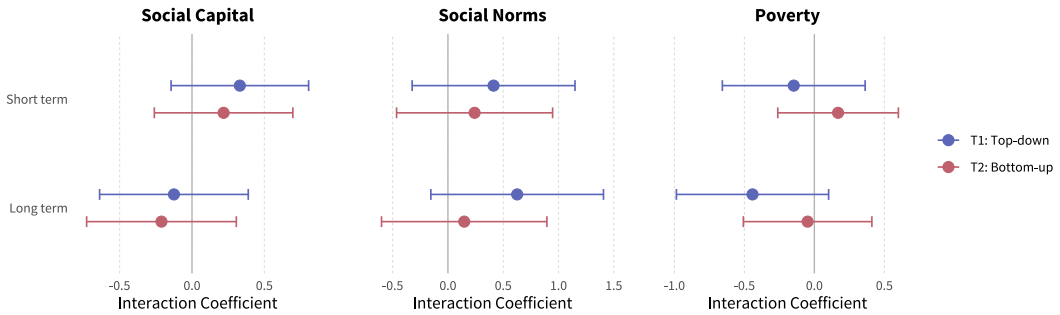
Notes: Interaction coefficients from raster-level regressions of contamination reduction on treatment indicators interacted with street type dummies. Large street includes major roads and connectors. Any street includes major roads, connectors, and residential streets. Specification includes baseline contamination and strata fixed effects. Standard errors clustered at the community level. Error bars represent 95% confidence intervals.

Figure A5: Effect heterogeneity by initial contamination, community size, and remoteness



Notes: Interaction coefficients from community-level regressions of contamination reduction on treatment indicators interacted with community characteristics. Specification includes baseline contamination and strata fixed effects. Error bars represent 95% confidence intervals.

Figure A6: Effect heterogeneity by social norms, social capital, and poverty



Notes: Interaction coefficients from community-level regressions of contamination reduction on treatment indicators interacted with baseline social characteristics. Specification includes baseline contamination and strata fixed effects. Error bars represent 95% confidence intervals.

A5 Additional survey and registry results

Table A8: Additional survey results for waste-related behaviors

	Short-term effects			*Long-term effects*		
	T1: TD	T2: BU	T2 - T1	T1: TD	T2: BU	T2 - T1
Waste disposal methods						
Disposes: deposit	-0.048	0.029	0.078***	-0.043	0.015	0.058
	(0.030)	(0.032)	(0.029)	(0.035)	(0.038)	(0.039)
Disposes: burn	-0.025	-0.138***	-0.113**	-0.016	-0.068*	-0.053
	(0.046)	(0.039)	(0.045)	(0.042)	(0.038)	(0.040)
Disposes: bury	-0.021	-0.002	0.019	0.009	-0.002	-0.011
	(0.013)	(0.015)	(0.013)	(0.014)	(0.015)	(0.015)
Disposes: truck	0.087	0.111**	0.024	0.036	0.053	0.017
	(0.054)	(0.050)	(0.051)	(0.057)	(0.055)	(0.057)
Disposes: dump	-0.004	-0.001	0.003	-0.002	-0.002	0.000
	(0.007)	(0.007)	(0.006)	(0.001)	(0.001)	(0.001)
Recycling behavior						
Recycles: plastic	-0.046	0.079*	0.125***	-0.054	0.056	0.110**
	(0.047)	(0.047)	(0.043)	(0.044)	(0.042)	(0.045)
Recycles: glass	0.019	0.036	0.017	-0.039	0.008	0.047
	(0.049)	(0.050)	(0.052)	(0.028)	(0.027)	(0.029)
Recycles: paper	0.014	0.056	0.043	-0.060**	0.001	0.061**
	(0.031)	(0.037)	(0.035)	(0.027)	(0.029)	(0.024)
Recycles: organic waste	0.109**	0.090**	-0.019	-0.010	-0.012	-0.002
	(0.043)	(0.041)	(0.037)	(0.048)	(0.051)	(0.048)
Recycling items (nr)	0.092	0.267**	0.175	-0.162	0.055	0.216**
	(0.113)	(0.125)	(0.111)	(0.098)	(0.102)	(0.095)
Willingness to pay for cleaning						
Pay for cleaning, me (log)	-0.110**	-0.109*	0.001	0.060	0.028	-0.031
	(0.056)	(0.061)	(0.050)	(0.068)	(0.061)	(0.062)
Pay for cleaning, others (log)	-0.085	-0.080	0.004	0.024	-0.026	-0.050
	(0.062)	(0.069)	(0.053)	(0.058)	(0.053)	(0.047)

Sample sizes are n = 2066 for the estimation of short-term effects and n = 1832 for long-term effects. Controls include strata fixed effects, sex, age, education (dummies), and the baseline value for the respective outcome. Standard errors are clustered at the community level. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$

Table A9: Survey regression results for social capital and other outcomes

	Short-term effects			*Long-term effects*		
	T1: TD	T2: BU	T2 - T1	T1: TD	T2: BU	T2 - T1
Trust in institutions						
Trust comm. leaders (sd)	0.035	0.104	0.069	-0.096	0.056	0.152**
	(0.103)	(0.093)	(0.106)	(0.076)	(0.065)	(0.064)
Trust municipal gov. (sd)	0.024	0.097	0.072	0.012	-0.061	-0.074
	(0.085)	(0.084)	(0.081)	(0.088)	(0.071)	(0.082)
Trust central gov. (sd)	0.045	0.130	0.085	-0.060	-0.044	0.016
	(0.114)	(0.110)	(0.107)	(0.073)	(0.064)	(0.070)
Community participation						
Collective action hours (log)	-0.035	0.188***	0.224***	0.031	0.224***	0.194**
	(0.053)	(0.057)	(0.061)	(0.076)	(0.080)	(0.077)
No. of organizations (count)	0.022	0.056	0.034	-0.005	0.035	0.040
	(0.036)	(0.036)	(0.043)	(0.032)	(0.038)	(0.044)
Other outcomes						
Living conditions (sd)	-0.141**	-0.094	0.046	-0.075	0.029	0.103
	(0.059)	(0.066)	(0.068)	(0.059)	(0.053)	(0.063)
Security (sd)	-0.015	0.053	0.068	-0.035	-0.074	-0.040
	(0.081)	(0.086)	(0.085)	(0.097)	(0.088)	(0.109)
Altruism (%)	-0.014	-0.001	0.013	0.013	0.017	0.004
	(0.019)	(0.020)	(0.020)	(0.017)	(0.015)	(0.015)

Sample sizes are n = 2066 for the estimation of short-term effects and n = 1832 for long-term effects. Controls include strata fixed effects, sex, age, education (dummies), and the baseline value for the respective outcome. Standard errors are clustered at the community level. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$

Table A10: Survey results for community activities

	Short-term effects			*Long-term effects*		
	Control	T1: TD	T2: BU	Control	T1: TD	T2: BU
Activities observed						
Community meeting	0.29	0.35	0.70***	0.24	0.21	0.40***
Session or workshop	0.06	0.13**	0.48***	0.01	0.02	0.05*
Cleaning	0.26	0.55***	0.74***	0.36	0.45	0.59***
None	0.55	0.38***	0.15***	0.57	0.50	0.31***
Activities participated						
Community meeting	0.16	0.15	0.42***	0.18	0.16	0.27**
Session or workshop	0.03	0.06	0.33***	0.01	0.01	0.04
Cleaning	0.21	0.34***	0.51***	0.30	0.36	0.47***
None	0.69	0.62	0.43***	0.64	0.61	0.47***
Perception						
Level of activities (sd)	0.00	0.16	0.81***	0.00	-0.06	0.20**
Waste management organization (sd)	0.00	0.25***	0.59***	0.00	0.00	0.22***
Frequency waste truck	1.86	2.44	2.26	1.79	2.19	2.20
Frequency waste truck usage	1.64	2.20	2.07	1.66	2.04	1.95
Frequency community cleaning	0.96	1.18	1.40	0.82	0.71	0.89

Sample sizes are n=2066 for the estimation of short-term effects and n=1832 for long-term effects. The displayed values are sample means per group. Stars indicate the significance level of each treatment effect compared to the control group. Standard errors were clustered at the community level. Controls are baseline education, sex, age, and strata fixed effects. $*p < 0.1$; $**p < 0.05$; $***p < 0.01$

Table A11: Community activities: Summary statistics

Activity	*During intervention*			*After intervention*		
	Completed activities	% with one or more	Participants	Completed activities	% with one or more	Participants
Capacity building	15.20	0.95	18.90	1.03	0.41	26.40
Collective action	3.40	0.92	18.80	1.64	0.67	16.80

The period for both during as well as after the intervention spans a total of 4 months. Intervention activities were recorded by facilitators. The post-intervention data was obtained through phone calls to the responsible person at each community. The number of participants is conditional on the activity taking place.

Table A12: Comparison of outcomes by poverty status

Outcome	Non-poor	Poor	Difference	P-value	N
Community-level comparison					
Contamination (trash count)	1.604	1.553	-0.051	0.143	120
Truck frequency (monthly count)	2.583	1.512	-1.071**	0.022	120
Truck frequency (log)	0.961	0.665	-0.296**	0.032	120
WTP for cleaning (USD, log)	0.458	0.244	-0.214***	0.007	120
Volunteering (%)	0.248	0.341	0.094***	0.002	120
Volunteering (log)	0.215	0.285	0.070***	0.002	120
Willing to contribute time to community (%)	0.912	0.946	0.034***	0.007	120
Willing to contribute money to community (%)	0.435	0.396	-0.038	0.277	120
Participated in cleanup (%)	0.322	0.381	0.059	0.184	120
Individual-level comparison					
WTP for cleaning (USD, log)	0.356	0.326	-0.031	0.625	2421
Volunteering (%)	0.298	0.283	-0.016	0.466	2421
Volunteering (log)	0.207	0.196	-0.011	0.473	2421
Willing to contribute time to community (%)	0.922	0.945	0.022**	0.038	2421
Willing to contribute money to community (%)	0.45	0.333	-0.118***	0.000	2421
Participated in cleanup (%)	0.344	0.38	0.035	0.235	2066

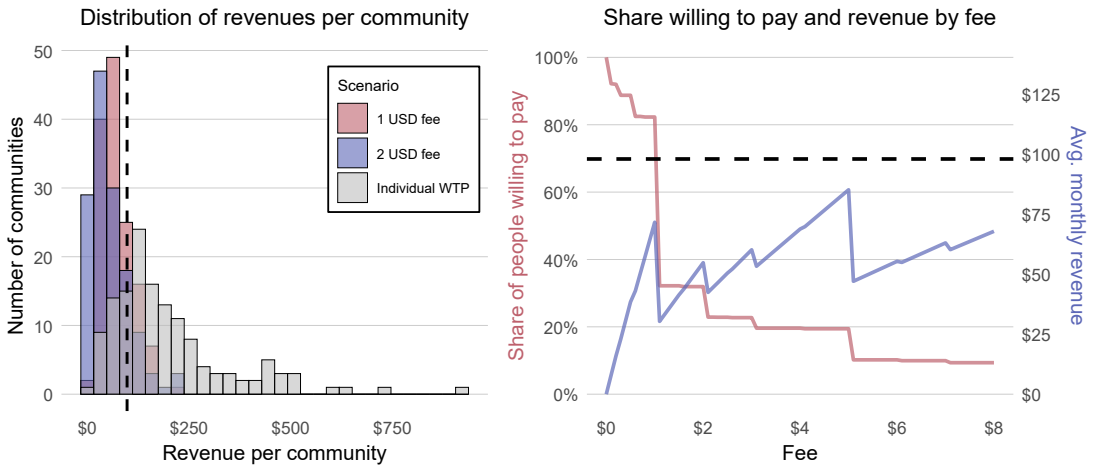
Poor and non-poor groups are defined by a median split of the baseline poverty assessment at the community level (upper panel) or individual level (lower panel); at the individual level the assessment is discrete, so the poor group comprises respondents rated strictly above the median and the two groups are not of equal size. Standard errors for individual-level outcomes are clustered at community level. Baseline values are used for all outcomes except truck frequencies and cleanup participation (post-intervention only). Willingness to contribute time/money should be interpreted with care as some enumerators misinterpreted these as either-or rather than two separate questions. *$p < 0.1$; **$p < 0.05$; ***$p < 0.01$

Table A13: Participation patterns by activity type

	Capacity building	Collective action
Participant shares		
Men (%)	16.8	15.2
Women (%)	53.9	51.6
Children (%)	29.3	33.1
Number of participants		
Poor communities (mean)	19.8	19.9
Non-poor communities (mean)	17.8	17.2
Poor communities (median)	19.5	17.5
Non-poor communities (median)	15.0	16.0

Results are based on registry data for bottom-up intervention communities. Communities are classified by median split of baseline poverty assessment.

Figure A7: Willingness to pay for waste collection (household level interpretation)



Notes: Left panel shows the distribution of total revenue per community under 1 USD and 2 USD monthly fees, plus a scenario where all households pay their stated WTP. Right panel shows the share willing to pay (left axis) and average monthly revenue (right axis) by fee level. Revenue calculations treat WTP as household-level contributions and assume survey respondents are representative of all households (total population divided by average household size of 3.1). WTP is winsorized at the 95th percentile. Dashed line indicates average operational costs (98 USD).

A6 Supplementary information on data and measurement instruments

Table A14: Coding of the Survey Questions

Variable	*Survey question*	*Possible answers*	*Computation*
Indices			
Social norms index	Composite index of: Descriptive, Injunctive, and Punishment norms	See subcomponents below	Standardized average
Descriptive norms	Sub-index of: Littering (me, reversed); Littering (others, reversed)	See individual questions below	Standardized average
Injunctive norms	Sub-index of: Littering is bad (me); Littering is bad (others)	See individual questions below	Standardized average
Punishment norms	Sub-index of: Punish littering (me); Punish littering (others)	See individual questions below	Standardized average
Social capital index	Composite index of: Networks, Trust, Organizations, and Collective action	See subcomponents below	Standardized average
Networks	Sub-index of: Strong ties; Weak ties	See individual questions below	Standardized average
Trust	Sub-index of: Trust; Trust (relative); Trust community leaders; Mutual help	See individual questions below	Standardized average
Organizations	Organizations	See individual question below	Standardized
Collective action	Voluntary work	See individual question below	Standardized
Waste-related behavior and social norms			
Perceived cleanliness	How do you evaluate the garbage contamination situation in your community?	Scale from 1 (Very clean) to 5 (Very dirty)	Standardized

Table A14: Coding of the Survey Questions

Variable	*Survey question*	*Possible answers*	*Computation*
Bothered by litter	Personally, how bothered are you by the trash in your community?	Scale from 1 (Not at all) to 5 (Very much)	Standardized
Appropriate disposal	In the past month, how has your household gotten rid of trash?	1: Trash truck; 2: Deposit; 3: Bury it; 4: Burn it; 5: Informal deposit, street	Percentage that used the trash truck or formal deposits
Recycling	In the past month, has your household separated any trash for recycling?	0: None of the below; 1: At least one of the below	Percentage
Recycling items	What types of garbage have been recycled?	0: None; 1: Plastic; 2: Glass; 3: Paper; 4: Organic waste	Number of different items
Pay for cleaning, me	Imagine if a service was hired in your community to clean the streets. How much would you be willing to contribute per month?	Decimal	Log
Pay for cleaning, others	On average, how much do you think a person in your community would be willing to contribute?	Decimal	Log
Littering, me	Being very, very honest, in the last month, have you ever thrown trash in the street?	0: No; 1: Yes	Percentage
Littering is bad, me	In your personal opinion, is it bad to litter on the street?	Scale from 1 (Not at all bad) to 5 (Very bad)	Standardized
Punish littering, me	If you observed someone in your community throwing trash in the street, what would you do?	0: Nothing, I do not want to get involved / it does not seem serious to me; 1: React with disapproval	Percentage of people reacting with disapproval

Table A14: Coding of the Survey Questions

Variable	*Survey question*	*Possible answers*	*Computation*
Littering, others	Out of every 10 people in your community, how many do you think have thrown trash in the street in the last month?	Integer	Percentage
Littering is bad, others	Out of every 10 people in your community, how many do you think believe it is wrong to litter in the street?	Integer	Percentage
Punish littering, others	Out of every 10 people in your community, how many do you think would react with a gesture of disapproval to a person throwing trash in the street?	Integer	Percentage
Social capital			
Strong ties	Q1: Approximately how many people live in your community?; Q2: Of these people, how many persons are close acquaintances (family, friends)?	Integers	Percentage (Q2/Q1)
Weak ties	Q1: Approximately how many people live in your community?; Q2: Of these people, how many persons are close acquaintances (family, friends)?; Q3: Of these people, how many persons are casual acquaintances?	Integers	Percentage ((Q2+Q3)/Q1)
Trust	People in your community will take advantage of each other	Scale from 1 (Completely disagree) to 5 (Completely agree)	Standardized, reverse coded
Trust (relative)	Compared to other people, do you trust people in your community more or less?	Scale from 1 (Much less) to 5 (Much more)	Standardized
Trust community leaders	How much do you trust your community leaders?	Scale from 1 (Not at all) to 5 (A lot)	Standardized

Table A14: Coding of the Survey Questions

Variable	*Survey question*	*Possible answers*	*Computation*
Mutual help	People in your community help each other	Scale from 1 (Completely disagree) to 5 (Completely agree)	Standardized
Organizations	Do you belong to one or more community organizations or groups (e.g. ADESCO, youth organization, collectives, etc.)?	0: No; 1: Yes	Percentage
Voluntary work	In the past month, have you participated in any type of volunteer work for the community?	0: No; 1: Yes	Percentage
Voluntary work hours	Hours spent on community volunteer work last month	Integer	
Other outcomes			
Living conditions	Compared to other communities, how do you evaluate the living conditions in your community?	Scale from 1 (Very bad) to 5 (Very good)	Standardized
Security	Compared to other communities, how do you evaluate the security situation in your community?	Scale from 1 (Very safe) to 5 (Very dangerous)	Standardized
Trust municipal government	How much do you trust municipal officials?	Scale from 1 (Not at all) to 5 (Very much)	Standardized
Trust central government	How much do you trust central government officials?	Scale from 1 (Not at all) to 5 (Very much)	Standardized

Table A14: Coding of the Survey Questions

Variable	*Survey question*	*Possible answers*	*Computation*
Altruism	There will be a lottery for 100 USD among the participants. The winner will have to decide how much of this money to keep and how much to donate to a family in need in the department (photos of the delivery would be sent).	Integers	Percentage of how much was donated
Socio-demographic variables			
Female	Gender	0: Male; 1: Female	
Age	Age	Integer	
Education	Highest level of education	1: None; 2: Incomplete primary; 3: Complete primary; 4: High school degree; 5: Technical; 6: University degree	Dummies for each level
Poverty	What is the family's economic situation like? The family's poverty level was recorded by the enumerators based on pictures of potential housing conditions.	Scale of 1 (Not poor) to 5 (Very poor)	Standardized
Community size	Approximately how many people live in your community?	Integer	
Waste management activities			
Activities observed	In the last 4 months, have you heard of any activity related to the issue of garbage in your community?	Community meeting; Session or workshop; Cleaning; None	Dummies for each activity

Table A14: Coding of the Survey Questions

Variable	*Survey question*	*Possible answers*	*Computation*
Activities participated	In the last 4 months, have you participated in any activity related to the issue of garbage in your community?	Community meeting; Session or workshop; Cleaning; None	Dummies for each activity
Level of activities	In the last 4 months, do you think there were more or fewer activities than before regarding the issue of garbage in your community?	Scale from 1 (Much less) to 5 (Much more)	Standardized
Waste management organization	In your opinion, how organized is your community in relation to garbage management?	Scale from 1 (Not at all organized) to 5 (Perfectly organized)	Standardized
Frequency waste truck	In the last 4 months, how often has a toilet train arrived in your community?	1: Never; 2: Every 2 months; 3: Every month; 4: Every 2 weeks; 5: Every week; 6: Twice a week; 7: Every day	Frequency per month
Frequency waste truck usage	In the last 4 months, how often have you used the garbage train to dispose of your garbage?	1: Never; 2: Every 2 months; 3: Every month; 4: Every 2 weeks; 5: Every week; 6: Twice a week; 7: Every day	Frequency per month
Frequency community cleaning	In the last 4 months, how often has your community been cleaned?	1: Never; 2: Every 2 months; 3: Every month; 4: Every 2 weeks; 5: Every week; 6: Twice a week; 7: Every day	Frequency per month

A7 Theoretical background

Our mechanism analyses are based on a simple framework of how capacity-building interventions can facilitate bottom-up public goods provision.[19] We assume that individuals are willing to contribute to a public good as long as a certain fraction of the group does, and that they differ in these *thresholds for conditional cooperation*. A threshold of 0 corresponds to people who always cooperate, while a threshold of 1 indicates that someone never cooperates even if everyone else in the group does. Evidence from laboratory studies shows that these extreme types are in the minority, and that most people exhibit behavior consistent with varying degrees of conditional cooperation (Fischbacher et al., 2001). Individual thresholds may be determined by numerous factors, such as the importance the person places on the public good (i.e., preferences), the individual's pro-sociality, or his or her resources. As people usually cannot observe the actual number of contributors, they act based on their *beliefs* about it. This means that individuals will start contributing as soon as they believe that the proportion of contributors is higher than their personal threshold, and stop doing so if they think that this is no longer the case. For a given distribution of thresholds, multiple equilibria may thus be possible.[20] In a repeated game, we would expect self-reinforcing positive or negative dynamics, as people continually adjust their contributions based on the observed contributions of others until a stable equilibrium is reached (Berger, 2021; Berger et al., 2023).

Even when a socially more desirable equilibrium exists, attaining it often requires coordinated action. Take the example of a group of workers deciding whether to go on strike. If most people are willing to participate as long as most others do so too, the strike can only take place if the group coordinates to act simultaneously. Ample research shows that allowing people to communicate with each other increases the chance of reaching a stable high-level equilibrium (Chaudhuri, 2011). Following Cowen (1992) and Dahlman (1979), we thus assume that coordinating collective action entails *transaction costs*. The magnitude of these costs depends on how well people know and trust each other, and on the institutions they set up to facilitate cooperation. This idea is reflected in the notion of social capital, commonly understood as "the norms and networks that facilitate collective action"

[19] For simplicity, we limit ourselves to the extensive contribution margin (i.e., whether people contribute). Yet, a very similar case can be made for the intensive contribution margin (i.e., how much people contribute).

[20] Consider a community where 40 percent of individuals will contribute as long as at least 30 percent of the population contributes, and 60 percent contribute as long as at least 80 percent contribute. In this case, three stable equilibria could be reached: one where no one contributes, one where 40 percent contribute, and one where everyone contributes.

(Woolcock, 2001, p. 9). Thus, at higher levels of social capital, members of a group are more likely to succeed in organizing to collectively provide or protect public goods (e.g., Anderson et al., 2004). A related idea concerns the concept of leadership. In most real-world scenarios, transaction costs are not perfectly divisible, meaning that a single individual (or a small group of individuals) must bear a large portion of these costs. The presence of a committed leader (or leadership team) should thus be critical for a group to overcome organizational constraints to collective action. This is in line with extensive empirical evidence documenting the importance of leadership for collective action and the provision of public goods (e.g., Glowacki and von Rueden, 2015; Sahin et al., 2015).[21]

Therefore, we conclude that bottom-up provision succeeds if the distribution of contribution thresholds allows for a high-level equilibrium and – in cases where coordination is required – if the community is sufficiently organized to coordinate collective action so that this equilibrium can be reached.

Based on this framework, we distinguish two basic mechanisms through which capacity-building interventions could facilitate bottom-up public goods provision. First, they can alleviate *informational constraints*. If individuals underestimate the share of others contributing to a public good, getting people to talk about the problem can eliminate these misconceptions. By raising awareness and conveying information on the public good, such interventions could also directly alter the distribution of thresholds, as people begin to care more about the problem or become more confident about their ability to address it. This could induce shifts in individual behaviors and *social norms*, allowing the community to reach higher levels of public goods provision. This behavioral mechanism is best represented as a continuous public good problem, meaning that the public good can be provided to a certain degree rather than either fully provided or not at all.[22] In the context of our waste intervention, this would be reflected in changes in behaviors and social norms related that prevent waste accumulation (i.e., how much waste ends up on

[21]Note that in some cases, the institutions groups set up to facilitate cooperation can also be seen as modifying the thresholds themselves (rather than lowering transaction costs). For example, if groups devise means of punishing defector, this would change people's thresholds for conditional cooperation. The same can be said if individuals get inspired by a charismatic leader (Jack and Recalde, 2015). For simplicity, we abstract from this alternative conceptualization, and view the organization of the collective action as a second-order public goods problem related to the bearing of transaction costs.

[22]The continuous characterization refers to the fact that each additional person contributing (e.g., if someone stops littering) improves provision. This does not imply that every level of contribution constitutes a stable equilibrium. As discussed above, threshold dynamics in conditional cooperation can create tipping points where social norms shift discontinuously, meaning only certain contribution levels correspond to stable equilibria.

the streets in the first place).

A second possible mechanism is related to *organizational constraints* and thus to the transaction costs that effective coordination entails. By bringing people together and encouraging them to set up organizational structures, capacity-building interventions could build *social capital* and local leadership, thereby facilitating coordination. Here, changes occur through coordinated collective action rather than individual behavior. This mechanism is typically conceptualized as a threshold public good problem, where the public good is either provided if coordination succeeds or not provided at all. In the context of waste management, this corresponds to organizing collective waste removal systems such as community cleanups.

While the main focus of our study is on comparing the effectiveness of capacity building for bottom-up provision with a more traditional top-down intervention, we will use this theoretical framework to make sense of patterns in our data, thereby contributing to a better understanding of why capacity-building community initiatives may work and where they may fail.

A8 Literature review

Table A15: Literature Review on Bottom-Up Strategies

Study	*Study type*	*Description and results*
Meta studies of community-based development (CBD) programs		
Casey 2018	Meta study on evolution of CBD and CBD RCTs.	A synthesis of seven CBD RCTs shows that CBD effectively delivers public goods and some economic benefits at a low cost in challenging environments. However, it does not seem to lead to lasting transformations in local decision-making or empowerment of the poor. This raises the question of how much participation is necessary to preserve the benefits of decentralization while minimizing the time costs imposed on impoverished communities.
Mansuri and Rao 2012	Meta study proposing general concept of CBD based on literature from different fields.	The report discusses the history of participatory development and presents a framework for understanding participatory development, emphasizing the concept of "civil society failure" and its interaction with government and market failures. It is based on literature from anthropology, economics, political science and sociology. Evidence on key development outcomes, public service delivery and quality, but also on issues related to CBD is reviewed. The report also discusses World Bank-funded projects, emphasizing the importance of local context, as well as effective monitoring and evaluation for successful outcomes.
Evaluations of community-based development(CBD) programs		
Arcand 2008	IV study with panel data on 71 villages with 756 households in Senegal.	This paper investigates the impact of a national CBD program on access to basic services, household expenditures, and child wellbeing. The program had a positive effect on villagers' access to clean water and health services, as well as on child malnutrition. Completed income-generating agricultural infrastructure projects and improved primary education significantly increased household expenditures per capita, while health and hydraulic projects did not.

Table A15: Literature Review on Bottom-Up Strategies

Study	*Study type*	*Description and results*
Avdeenko and Gilligan 2015	RCT with 576 households in 24 communities, and 475 lab-in-the-field subjects.	The intervention had no impact on networks and social norms, but it increased people's involvement in civic activities and local governance. Therefore, the authors attribute the increase in citizen participation not to the growth of social capital, but to the greater openness of the local governments.
Beath et al. 2013	RCT with 500 villages in Afghanistan	The RCT examines the impact of a CBD program that requires female participation on several outcomes related to women's empowerment. Positive effects on women's participation in economic, social, and political activities are reported. However, no impacts on gender roles or family decision-making are found.
Casey et al. 2012	RCT with 2,832 households in 236 villages in the Republic of Sierra Leone.	The study evaluates a CBD program aiming to make local institutions more democratic and egalitarian by imposing participation requirements for marginalized groups. The program had positive short-term effects on local public services and economic outcomes. However, it did not result in sustained impacts on collective action, decision-making, or the involvement of marginalized groups, indicating that the intervention did not durably reshape local institutions.
Desai and Olofsgård 2019	RCT combined with behavioral experiment with 80 villages in India.	The "self-help" groups established in treatment villages significantly improved people's access to and the quality of certain public goods, especially water, due to better information through the groups, stronger community engagement and reduced coordination costs. The behavioral experiment 4 years after the RCT revealed that cooperative norms are stronger in villages that had self-help groups.
Fearon et al. 2009	RCT with 83 communities in Liberia	The study evaluates the impact of a community-driven (post-war) reconstruction project on social cohesion, as measured by an anonymous public goods game. Contributions were significantly higher in the treated communities, with a 9 percent increase in funds raised for a community-selected public good.

Table A15: Literature Review on Bottom-Up Strategies

Study	*Study type*	*Description and results*
Humphreys et al. 2019	RCT with 1,250 communities in the Congo	The study evaluates the impact of a community-driven reconstruction program on democratic governance. Behavior in an unconditional cash transfer program is used to assess whether the intervention had an impact on elite capture. No effects are found.
Labonne and Chase 2011	DiD with 2,100 households in 135 communities in the Philippines.	Using difference-in-differences (DiD) and propensity score matching, the study evaluates a CBD program where communities competed for grants for infrastructure investments. The program increased the participation in village meetings and the frequency of interactions between local officials and village officials, but had a negative impact on collective action.
Nguyen and Rieger 2017	RDD with 1,300 communes in Morocco	The study assesses the impact of a CBD initiative on social capital, employing a regression discontinuity design (RDD) based on the program's poverty selection threshold. The program increased contributions in a public goods game, but had no effect on altruism and a negative effect on trust.
Saguin 2018	DiD based on surveys in 16 municipalities in the Philippines	The "KALAHI-CIDSS" CBD program was found to increase the incomes of poor households. However, it did not improve outcomes such as solidarity and trust. In addition, poor households are underrepresented in village assemblies, with declining participation over time.
Van der Windt and Mvukiyehe 2020	RCT with 1,250 villages in the Republic of Congo.	The study assesses the long-term impact of a CBD initiative 8 years after its launch. The program had a lasting impact on infrastructure quality (e.g., of schools or hospitals), but no effects on other dimensions of service delivery, on economic welfare, and on local institutions (e.g., governance, social cohesion, or female empowerment) were found.

Related studies

Table A15: Literature Review on Bottom-Up Strategies

Study	*Study type*	*Description and results*
Banerjee et al. 2010	RCT with three interventions in India.	This paper examines if citizen involvement can shape public service provision in education. Three interventions were evaluated: (i) providing information on public school organization, (ii) introducing citizens to a simple monitoring tool for their local school, and (iii) training volunteers to hold reading camps in order to improve literacy knowledge. Information and monitoring did not improve outcomes, but the volunteer-led reading camps did.
Björkman and Svensson 2009	RCT with 50 public dispensaries in Uganda.	The intervention aimed at encouraging community engagement in monitoring health services and holding local health providers accountable for their performance. To this end, community members developed village action plans together with the health care providers. One year after the intervention, treatment communities exhibited greater involvement in monitoring providers, resulting in increased effort from health workers to serve the community as well as significant improvements in healthcare utilization and health outcomes.
Björkman Nyqvist et al. 2017	Follow-up of RCT in Björkman and Svensson (2009).	The authors evaluate the long-run impact (4 years) of the experiment in Björkman and Svensson (2009). Even with minimal follow-up, short-term enhancements in healthcare delivery and health outcomes were sustained over the long run. The results indicate that a lower-cost version of the treatment, which primarily aimed to boost participation without information on staff performance, did not influence the quality of care or health outcomes both in the short and in the in the longer run.
Duflo et al. 2015	RCT with 70 schools in Kenya.	The study evaluates a program in Kenya where parents and the school committees of randomly selected schools received (i) funding or (ii) funding and a short School-Based Management (SBM) empowerment training to hire an additional teacher, outside normal Ministry of Education civil-service channels. Centrally hired civil-service teachers in schools receiving only funding endogenously reduced their effort and captured rents for their families by getting relatives the contract teacher positions. The SBM program cut by half both the reduction in the regular teacher effort in response to the program and the fraction of contract teachers who were relatives of regular teachers.

Table A15: Literature Review on Bottom-Up Strategies

Study	*Study type*	*Description and results*
Olken 2007	RCT in 608 villages in Indonesia.	The paper evaluates different interventions aiming at reducing corruption, measured by missing expenditures, in village road projects in Indonesia. Results show that increased government audits significantly reduce missing expenditures. In contrast, enhancing grassroots monitoring had limited impacts on corruption.
Raffler et al. 2019	RCT with 376 health care centers and 14,609 households in rural Uganda.	The authors evaluate a large-scale information intervention aiming to improve bottom-up monitoring of health service delivery. The study finds only modest positive effects of citizen monitoring on service quality and patient satisfaction, and no effects on utilization and health outcomes such as child mortality.

Table A16: Literature Review on Waste Management

Study	*Study type*	*Description and results*
Evaluations of waste interventions		
Bateson et al. 2013	Field experiment with 620 bicycle riders on Newcastle university campus, UK.	This study tests if displaying images of "watching eyes" causes people to litter less and if a potential effect depends on the cleanliness of the environment. People were more likely to litter in dirty environments, but the watching eyes only had an effect when many people were around, and this effect does not depend on the amount of litter in the environment.
Castaldi et al. 2021	RCT in 8 beach resorts in Italy.	The resorts were randomly assigned to 3 groups: (i) free portable ashtrays, (ii) free portable ashtrays and anti-littering message, and (iii) control. Results show a reduction in daily litter (cigarette butts in sand on day/costumers): -10% to -12% for the ashtray group; -7% to -10% for the ashtray + message group.
Cialdini et al. 1990	5 field experiments in different public spaces with 127–484 observations.	The authors argue that injunctive and descriptive norms must be separated to understand littering behavior since behavior changes only in accordance with the more salient type. In their experiments, they find that littering increases in littered environments, and even more so when someone is observed littering. Conversely, littering decreases when someone is observed littering into a very clean environment. Men are more likely to litter than women across different settings.
Dur and Vollaard 2015	Field experiment with 4,000 households in the Netherlands.	This paper studies littering behavior and free-riding mechanisms related to public services. In a randomly assigned part of residential area, the frequency of cleaning around the garbage containers is drastically reduced from daily cleanups to 2-3 times a week during a 3-month period. Removing the morning cleanup increased the presence of litter in the early afternoon (11% to 27%). Litter accumulation around the garbage disposal increased (from 20% to 75%). Telephone appointments for retrieval of large trash increased, meaning that some people started to clean up more by themselves. The effects persisted at least one month after the treatment ended.

Table A16: Literature Review on Waste Management

Study	*Study type*	*Description and results*
Lewis et al. 2009	Nationwide survey on littering attitude and field study in cinemas in the UK.	The survey revealed personal differences in acceptance and justification of littering depending on the age group, rural/urban living environment, smoker/non-smoker, feeling connected/unconnected to community. Also, missing infrastructure was identified as a cause for littering. In the cinema field study, leaflets with a (i) control message unrelated to littering, (ii) polite anti-littering message, or (iii) direct anti-littering message were distributed, and it was observed how much litter was left behind. People in the control group littered more than people that were politely or directly asked not to.
Liu and Sibley 2004	Field study in a public space in New Zealand with over 3,000 observations.	In a first sub-study, littering attitudes were observed during 3 weeks and the people who disposed of waste (correctly and incorrectly) were interviewed. In the second week, a banner with an anti-littering message was added. People were found to litter less in crowded public spaces compared to less-crowded public places. The banner did not change littering behavior. In a second sub-study, bins and ashtrays were installed, and found to reduce littering by 64% without changing attitudes towards littering.
Nepal et al. 2023	RCT with 75 treatment and 75 control communities in Nepal.	The study evaluates a low-cost treatment to improve municipal solid waste management: Providing information to households and installing waste bins on the streets. Perceived cleanliness in treatment communities increased by 25% at midline (3 months after installation) and 43% at endline (9 months after installation). Giving household waste to collectors increased by 13% at midline and 9% at endline while there was no statistically significant change in at-source waste segregation.
Ramos and Torgler 2012	Field study with 98 observations in Australia.	The authors test the broken-window-theory in a field study, which was conducted over 6 days in university common rooms, alternating between an orderly and a disorderly environment. 59% of participants littered in the disorderly room, compared to 18% in the clean room. Multivariate analysis shows that the disorder variable is always large and statistically significant. Older individuals and senior staff were more likely to litter.

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Study	*Study type*	*Description and results*
Rangoni and Jager 2017	Simulation in an agent-based model with 100 simulated pedestrians.	The goal of the simulation is to evaluate how social influence may cause a transition from a clean to a littered environment in 3 situations: (i) no trash bins; (ii) trash bins which can get full, and (iii) adding cleaners who can pick up litter and empty bins. For the parameterization of the model, data from a field study is used. The simulations suggest that litter does not grow linearly. Furthermore, a dynamic cleaning regime is cheaper and more effective than pre-determined regimes.
Sagebiel et al. 2020	Field experiment with 200 observations on university benches in Germany.	To test the broken-window theory in the context of littering cigarette butts, two types of environment were prepared: (i) clean environment in which all cigarette butts were removed around the benches; (ii) dirty environment in which 25 cigarette butts were placed around each bench. The authors conclude that increased cleaning effort reduces littering a little, but the effect might be too small to justify additional cleaning costs.
Schultz 1999	Field experiment with 605 residents of single-family dwellings in the US.	The study aims to find out if a plea alone or accompanied by (i) information, (ii) neighbor feedback, or (iii) household feedback increases proper waste disposal. Results show that feedback targeting personal or social norms increased the proportion of people recycling and the amount of recycled materials while not changing the level of contamination through littering. The author argues that a link between norm activation and behavior change exists.
Sheely 2013	RCT with 36 communities in Kenya (based on qualitative study).	This study aims to explain variation in maintaining a clean environment through interactions between government and community institutions. Communities are randomly assigned to 4 experimental groups: (i) collective action to organize cleanups promoted by a local NGO, (ii) collective action and punishment for littering by government chiefs, (iii) collective action and punishment for littering by traditional elders, and (iv) a control group. The author finds that communities with no formal punishment for littering experienced a sustained reduction in littering behavior and an increase in the frequency of public cleanups. Communities in which government administrators or traditional leaders punished littering experienced short-term reductions in littering that were not sustained.

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Study	*Study type*	*Description and results*
Other waste studies		
Chitotombe 2014	Interviews in Zimbabwe and literature review.	The unavailability of bins, socio-cultural consumption styles in particular related to fast foods, illegal display of posters in the streets, and abandoned motor vehicles are mentioned as problems in Zimbabwe. Anti-littering campaigns have shown little success in the past. The interviews revealed that bins were not used even if available and that communities are reluctant to participate in cleanups. Also, the study shows that language barriers and political inefficiencies impede proper waste management.
Nkwocha and Okeoma 2009	Interviews of 6,000 individuals in 6 geo-political zones in Nigeria.	Littering is very common in Nigeria. Reasons given by respondents included lack of bins or long distances to dumpsites, inefficiency of local authorities in keeping public spaces clean, missing legislation against littering, convenience, and ignorance of the environmental and health consequences of littering. Low levels of education were highly correlated with littering.
Tanyanyiwa 2015	Interviews with residents and street workers in Zimbabwe.	Reasons for the high littering levels are identified as: missing sense of ownership of public areas, the belief that someone else will clean up, and that littering is tolerated. Suggested ways to reduce litter include the provision of dedicated recycling bins, a volunteer environmental police force, and the establishment of a coordinated waste management system.
Torgler et al. 2009	Analysis of over 30,000 respondents of the European Value Survey (EVS).	Using EVS data on basic values and beliefs of people in Europe, the authors find a positive, albeit small, relationship between how people perceive environmental cooperation (public littering) and their voluntary environmental morale.